

Attitude Sensing with Backscatter Arrays: Identifiability, Intrinsic Bounds, and Geometry Design

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Abstract—Point-target sensing describes an object by range, angle, and Doppler, none of which reveals the object’s attitude or angular velocity. A rigid body carrying a calibrated backscatter array encodes both states in its echo phases, which one illuminator and several mutually incoherent receive apertures can read. This paper develops the identifiability theory and the intrinsic performance limits of this encoding, and derives the geometry design laws that follow from both. The attitude enters every wrapped echo phase as a bilinear form between an aperture-side effective viewing direction and a target-side baseline, alongside translational, oscillator, and gain terms. With the viewing directions spanning d_u dimensions and the baselines d_b , recovering an unconstrained attitude’s nine entries needs $d_u d_b = 9$, and we prove that the rotation constraint relaxes this to $d_u + d_b \geq 5$, which is necessary and sufficient for identifiability and equivalent to a nonsingular Fisher information at every attitude. For geometries below this threshold, we provide a complete catalog of ambiguities. The Fisher information factors into an aperture-side matrix and the attitude-conjugated target-side spread, so the intrinsic Cramér–Rao bound depends on the attitude being estimated unless the spread is isotropic. One conjugate difference across antennas removes the three nuisance terms without loss of attitude information, and a grid search over the rotation group with integer least squares at every node resolves the ambiguities under a deterministic covering condition. The minimum spanning tree minimizes the integer search at a fixed baseline budget, and the worst-axis criterion is maximized at a bistatic angle of ninety degrees, with a closed-form ten-percent performance sector. Chip-level simulations confirm threshold, bounds, and design laws.

Index Terms—Attitude estimation, angular velocity, backscatter array, bistatic geometry, identifiability, intrinsic Cramér–Rao bound, integer ambiguity resolution, rotation group.

I. INTRODUCTION

Integrated sensing and communications (ISAC) turns the cellular infrastructure into a sensor, but its radar-inherited point model omits the states that matter for the platforms it serves [1]. In that model an object is a point with a range, an angle, and a Doppler shift. A vehicle at an intersection is described by its heading and yaw rate, and a drone on approach by its attitude and angular velocity [2]. Neither state is observable under a point-target model, because a rotation about the center of mass leaves the point unchanged. Attitude evolves on $SO(3)$, so its estimation, identifiability, and performance

limits call for an intrinsic treatment rather than Euclidean coordinates [3], [4]. The deployment considered here has one illuminator and several receive apertures. This bistatic mode is increasingly considered in ISAC standardization and system studies, since separating transmission from reception removes the self-interference of a monostatic node [5]–[7]. The target is cooperative in one respect only. It carries a passive backscatter array of known layout and applies known load codes, with hardware complexity and power consumption comparable to those of passive RFID devices [8], [9].

Existing methods can be grouped into five classes according to the type of observation, which in turn determines the appropriate estimator. Complete observations of body-frame directions pose Wahba’s problem, solved in closed form or by maximum likelihood [10], [11]. Scalar projections of a baseline onto known directions, observed with an integer ambiguity, arise as carrier-phase differences between antennas. They are solved by integer least squares under the orthonormality constraint [12]–[14], or by an ambiguity-function search over the parameter itself [15]. Ranges, delays, angles, or Doppler shifts between body-mounted nodes and external anchors are solved by constrained least squares, semidefinite relaxation, or message passing [2], [16]–[20]. Angles and delays of resolvable propagation paths at a user array are solved by maximum likelihood on the manifold [21], [22]. Phase differences across the antennas of a backscatter array under a single link are read by an orientation spectrum, a polarization-sensitive phase model, direction finding, or designed response codes [23]–[26]. The second and the fifth classes observe the same quantity, a body-frame baseline projected onto a viewing direction, and so does this paper.

For these observations, the existing results on identifiability and performance limits can be organized into four strands of work. With complete vector observations, two noncollinear vectors identify the attitude, and the spectrum of the information matrix does not depend on the attitude [3], [11]. Performance limits on the rotation group are stated by the intrinsic Cramér–Rao bound. It measures the error in the tangent space and coincides with the constrained bound under the orthonormality constraint [4], [27]. Rigid-body localization uses this bound as its benchmark and evaluates it numerically for a given geometry [16], [28]. For scalar projections, a two-step algorithm shows that three dimensions on one side and two noncollinear vectors on the other suffice. It also gives the information in a pairwise form [29]. A single baseline

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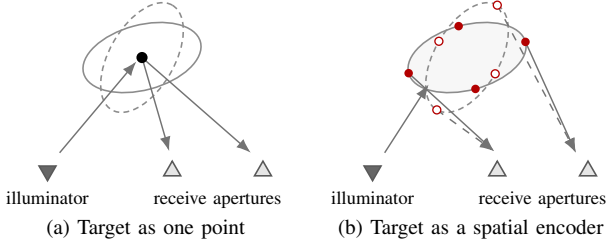


Fig. 1. Two attitudes of the same body (solid, dashed) under the point-target model and with a known backscatter array. (a) One echo for both attitudes, with identical range, angle, and Doppler. (b) One phase pattern per attitude, with distinct antenna phase differences.

determines two attitude angles only [30]. The rotation about a single measured direction is unobservable, and time-varying excitation recovers it [31]. Orientation estimation from several stations needs at least two of them in numerical results [32]. For a bilinear model without the rotation constraint, identifiability is a product-type count of the two dimensions [33]. Each of these results fixes one side, or evaluates the bound for one geometry at a time.

In Fig. 1, one illuminator probes a cooperative body carrying an array of known layout, and several receive apertures collect the echoes. At each aperture, the phase of each antenna depends bilinearly on the aperture-side effective viewing direction and the target-side body-frame baseline. The effective viewing direction is the sum of the unit vectors from the illuminator and from that aperture toward the target. Under the first-order target-side far-field model, the bistatic geometry enters only through this direction. The span of the viewing directions and the span of the baselines are the two sides of the observation. The observation is built from the load-coded echo, in which the translational phase and an independent oscillator phase per aperture enter as column-common terms and the gains are calibrated. Every phase is observed modulo 2π , and no carrier coherence among the apertures is assumed. Viewing directions may come from one large array, from separate receive apertures, or from target motion.

A. Motivations and Contributions

The bilinear form places the two spans on an equal footing and raises two questions that the results above do not reach. Since the attitude enters only through products of a viewing direction and a baseline, whether the phases determine it can depend on the two spans only jointly. The first question is what joint condition on the spans makes the attitude identifiable, and which attitudes remain indistinguishable when the condition fails. The Fisher information is built from the same products. The second question is how it depends on the two geometries and on the attitude itself, and whether it separates into an aperture-side factor and a target-side factor. That separation decides whether the two sides can be designed apart.

This paper answers both questions, and the answers carry over into a design. The contributions are as follows:

- We prove that, given the integer ambiguities, the attitude is identifiable if and only if $d_u + d_b \geq 5$, with d_u and d_b the dimensions of the two spans, whereas nine free entries

would need $d_u d_b = 9$. The same count is equivalent to a nonsingular Fisher information at every attitude. Below it we catalog every ambiguity, including the geometries where the Fisher information is nonsingular while two attitudes fit the same phases.

- We derive the intrinsic Fisher information in factored form, an aperture-side matrix acting on the target-side spread conjugated by the attitude. The bound therefore depends on the attitude being estimated, and isotropy of the spread is the only condition that removes the dependence. A one-dimensional span on either side loses a rotation axis that the other side cannot restore. Two pulses bound the angular velocity at twice the attitude bound over the squared pulse interval.
- We show that the attitude information resides in the phase differences across the array. The common phases can be discarded, and a network without carrier coherence attains the bound of a coherent one. A grid search over $\mathcal{SO}(3)$ with integer least squares at every node, sized by a closed-form covering condition, attains that bound at high SNR with the integers resolved.
- We turn the factorization into design laws. The target-side spread is independent of the reference antenna, and its trace sets the accuracy level while its isotropy sets the attitude independence. The minimum spanning tree minimizes the integer search at a fixed baseline budget. The bistatic angle trades viewing diversity against phase sensitivity, so the receive apertures belong in a sector about its optimum rather than at a point.

Simulations from the chip level up reproduce the identifiability threshold and the ambiguity catalog, and the estimator attains the bound once the integers are resolved. Over a deployment field the bound maps follow the design laws, with aperture placement deciding coverage and target-side isotropy deciding how far the attitude can move the bound.

B. Organization and Notations

The remainder of the paper is organized as follows. Section II presents the system model and the estimation problem, Section III develops the estimators, Section IV presents the identifiability and performance-limit results, Section V derives the design laws, Section VI reports simulations, and Section VII concludes the paper.

Notation: Boldface lowercase and uppercase letters denote column vectors and matrices. $(\cdot)^T$, $(\cdot)^H$, and $(\cdot)^\dagger$ denote transpose, conjugate transpose, and Moore–Penrose inverse, $\|\cdot\|$ the Euclidean norm, $\otimes$ and $\odot$ the Kronecker and Hadamard products, and $\text{vec}(\cdot)$, $\text{diag}(\cdot)$, $\text{blkdiag}(\cdot)$, $\text{tr}(\cdot)$, $\text{rank}(\cdot)$, and $\text{span}(\cdot)$ vectorization, diagonal and block-diagonal concatenation, trace, rank, and span. $\mathbf{I}_n$, $\mathbf{1}_n$, and $\mathbf{0}$ are the identity matrix, the all-ones vector, and the zero vector of conforming size, and $\mathbb{E}[\cdot]$ is expectation. $\lfloor \cdot \rfloor$ is elementwise rounding to the nearest integer. For $\mathbf{x} \in \mathbb{R}^3$, $[\mathbf{x}]_\times$ is the skew-symmetric matrix with $[\mathbf{x}]_\times \mathbf{y} = \mathbf{x} \times \mathbf{y}$, $\text{Exp}(\boldsymbol{\theta}) = \exp([\boldsymbol{\theta}]_\times)$, and Log is its inverse on the principal branch. A dot denotes the time derivative, and $j = \sqrt{-1}$.

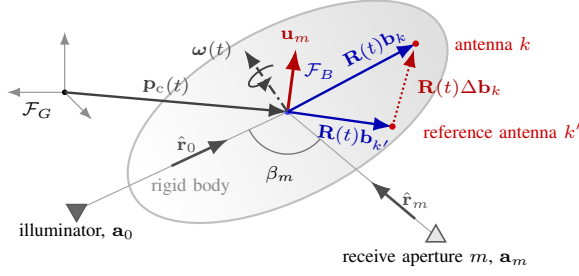


Fig. 2. System geometry of the Backscatter-MIMO target: the translation $\mathbf{p}_c(t)$, the rotated antenna positions $\mathbf{R}(t)\mathbf{b}_k$ and $\mathbf{R}(t)\mathbf{b}_{k'}$, the differencing baseline $\mathbf{R}(t)\Delta\mathbf{b}_k$, the angular velocity $\boldsymbol{\omega}(t)$, and the effective viewing direction $\mathbf{u}_m = \hat{\mathbf{r}}_0 + \hat{\mathbf{r}}_m$ of (9) along the bisector of the bistatic angle β_m .

II. SYSTEM MODEL AND PROBLEM FORMULATION

This section establishes that the rigid-body attitude enters the backscattered echo only through the effective viewing directions. It defines the single-illuminator multi-aperture geometry and the $\mathcal{SO}(3)$ perturbation convention, introduces the two-time-scale coded echo model, identifies the effective viewing directions, and states the estimation problem.

A. Geometry, Kinematics, and Lie-Algebra Convention

A single illuminator at known phase-center position $\mathbf{a}_0 \in \mathbb{R}^3$ transmits the probing signal, and M receive apertures at known positions $\{\mathbf{a}_m\}_{m=1}^M$ acquire the backscattered returns. An aperture is one receive antenna with its own downconversion chain. The elements of a large receive array and the antennas of separated reception points enter alike. Their local-oscillator (LO) phases may coincide or differ, and no relation among them is assumed or exploited.

The target is a cooperative rigid body carrying a Backscatter-MIMO array of K antennas. Antenna k sits at the known, time-invariant body-frame position $\mathbf{b}_k \in \mathbb{R}^3$ of the target frame $\mathcal{F}_B$. Let $\mathbf{p}_c(t) \in \mathbb{R}^3$ denote the target center-of-mass (CoM) position in the global coordinate frame $\mathcal{F}_G$, and let $\mathbf{R}(t) \in \mathcal{SO}(3) = \{\mathbf{R} \in \mathbb{R}^{3 \times 3} \mid \mathbf{R}^T \mathbf{R} = \mathbf{I}_3, \det(\mathbf{R}) = 1\}$ denote the attitude that maps body-frame vectors to $\mathcal{F}_G$. The global position of the k -th Backscatter-MIMO antenna is

$$\mathbf{p}_k(t) = \mathbf{p}_c(t) + \mathbf{R}(t)\mathbf{b}_k. \quad (1)$$

The attitude $\mathbf{R}(t)$ and the angular velocity $\boldsymbol{\omega}(t)$ are the rigid-body states of interest. Translation $\mathbf{p}_c(t)$ is a nuisance state and is never estimated. Local attitude perturbations use the left-perturbation convention

$$\mathbf{R}(\boldsymbol{\vartheta}) = \text{Exp}(\boldsymbol{\vartheta})\mathbf{R}_0, \quad (2)$$

where $\mathbf{R}_0$ is the nominal attitude and $\boldsymbol{\vartheta} \in \mathbb{R}^3$ is a tangent-space perturbation expressed in $\mathcal{F}_G$. The attitude evolves as

$$\dot{\mathbf{R}}(t) = [\boldsymbol{\omega}(t)]_{\times} \mathbf{R}(t), \quad (3)$$

where $\boldsymbol{\omega}(t) \in \mathbb{R}^3$ is the angular velocity expressed in $\mathcal{F}_G$. Differentiating (1) gives

$$\dot{\mathbf{p}}_k(t) = \dot{\mathbf{p}}_c(t) + [\boldsymbol{\omega}(t)]_{\times} \mathbf{R}(t)\mathbf{b}_k, \quad (4)$$

whose first term is common to all target antennas and whose second is antenna dependent. Fig. 2 shows the geometry.

B. Backscatter-MIMO Signal Model

The two-way phase of every coded channel splits into two terms. One term is common to all target antennas of an aperture, and the other pairs that aperture's viewing direction with the antenna's body-frame position.

The sensing process is organized over two time scales. The slow-time index $p = 1, \dots, P$ captures inter-pulse rigid-body motion, with sampling time $t_p = t_0 + (p-1)T_{\text{PRI}}$. Here t_0 is the center time of the first pulse and T_{PRI} is the pulse repetition interval. We use the shorthand notation $\mathbf{p}_c^{(p)} = \mathbf{p}_c(t_p)$, $\mathbf{R}^{(p)} = \mathbf{R}(t_p)$, and $\boldsymbol{\omega}^{(p)} = \boldsymbol{\omega}(t_p)$. Under the stop-and-go approximation, the target state is treated as constant within each pulse and evolves only across pulses. When adjacent-pulse kinematics are used, we adopt the piecewise-constant angular-velocity model

$$\mathbf{R}^{(p)} = \text{Exp}\left(T_{\text{PRI}}\boldsymbol{\omega}^{(p)}\right)\mathbf{R}^{(p-1)}. \quad (5)$$

Within pulse p , the fast-time chip index $l = 1, \dots, L$ implements Backscatter-MIMO load modulation. During chip l , the k -th target antenna applies the reflection coefficient $c_{k,l}$ drawn from the coding matrix $\mathbf{C} = [c_{k,l}]_{k \in [1:K], l \in [1:L]} \in \mathbb{C}^{K \times L}$. This fast-time coding separates the target-side antenna responses after matched filtering. We assume $L \geq K+1$ so that K orthogonal non-DC codes fit in one pulse.

The illuminator transmits a continuous-wave probing signal. Its transmit pattern stays fixed over the pulse, so all fast-time structure comes from the load modulation. After downconversion with the LO of aperture m and chip-level integration, the baseband sample at chip l of pulse p is

$$y_m^{(p)}[l] = e^{j\psi_m^{(p)}} \sum_{k=1}^K h_{m,k} c_{k,l} e^{-j\phi_{m,k}^{(p)}} + \zeta_m^{(p)}[l] + n_{m,l}^{(p)}, \quad (6)$$

where $\psi_m^{(p)}$ is the LO phase of aperture m at pulse p and $\phi_{m,k}^{(p)}$ is the two-way propagation phase of (7) below. The gain $h_{m,k} \in \mathbb{C}$ is calibrated and quasi-static. It excludes the geometric phase and the LO phase, and covers the remaining illuminator-target-aperture path through antenna k and aperture m . The term $\zeta_m^{(p)}[l]$ collects environmental clutter and interference, and $n_{m,l}^{(p)} \sim \mathcal{CN}(0, \sigma^2)$ is circularly symmetric complex Gaussian chip-level receiver noise.

The environmental term splits as $\zeta_m^{(p)}[l] = z_m^{s,(p)} + \xi_m^{(p)}[l]$, with clutter amplitude $z_m^{s,(p)}$, constant over the L chips of pulse p , and zero-mean colored interference $\xi_m^{(p)}[l]$, which absorbs any faster clutter variation. Collecting the latter in $\boldsymbol{\xi}_m^{(p)} = [\xi_m^{(p)}[1], \dots, \xi_m^{(p)}[L]]^T$ gives the covariance $\mathbf{R}_{\xi}^{(m)} = \mathbb{E}[\boldsymbol{\xi}_m^{(p)}(\boldsymbol{\xi}_m^{(p)})^H]$.

The probing signal travels from the illuminator to Backscatter-MIMO antenna k and returns to receive aperture m , so the two-way propagation phase is

$$\phi_{m,k}^{(p)} = \kappa_{\lambda} \left(\|\mathbf{p}_k^{(p)} - \mathbf{a}_0\| + \|\mathbf{p}_k^{(p)} - \mathbf{a}_m\| \right), \quad (7)$$

where $\kappa_{\lambda} \triangleq 2\pi/\lambda$. This form is exact for continuous-wave probing from a single transmit phase center.

Let $r_0^{(p)}$ and $r_m^{(p)}$ be the ranges from the illuminator and from aperture m to the target CoM, with unit directions $\hat{\mathbf{r}}_0^{(p)}$ and $\hat{\mathbf{r}}_m^{(p)}$. Under the target-side far-field condition

$\min\{r_0^{(p)}, r_m^{(p)}\} \gg \max_k \|\mathbf{b}_k\|$, a first-order expansion of both distances in (7) gives

$$\begin{aligned} & \|\mathbf{p}_k^{(p)} - \mathbf{a}_0\| + \|\mathbf{p}_k^{(p)} - \mathbf{a}_m\| \\ & \approx r_0^{(p)} + r_m^{(p)} + (\hat{\mathbf{r}}_0^{(p)} + \hat{\mathbf{r}}_m^{(p)})^T \mathbf{R}^{(p)} \mathbf{b}_k, \end{aligned} \quad (8)$$

with error of order $O(\|\mathbf{b}_k\|^2 / \min\{r_0^{(p)}, r_m^{(p)}\})$. The approximation is imposed only over the target-side aperture. Both geometries enter through the sum of the two unit directions alone, the effective viewing direction of aperture m ,

$$\mathbf{u}_m^{(p)} \triangleq \hat{\mathbf{r}}_0^{(p)} + \hat{\mathbf{r}}_m^{(p)}. \quad (9)$$

Then (8) splits the two-way phase into

$$\phi_{m,k}^{(p)} = \phi_m^{\text{trans},(p)} + \phi_{m,k}^{\text{rot},(p)}, \quad (10)$$

where $\phi_m^{\text{trans},(p)} = \kappa_\lambda(r_0^{(p)} + r_m^{(p)})$ is the translational phase shared by all target antennas of one aperture, and $\phi_{m,k}^{\text{rot},(p)} = \kappa_\lambda(\mathbf{u}_m^{(p)})^T \mathbf{R}^{(p)} \mathbf{b}_k$ is the attitude-dependent Backscatter-MIMO spatial phase. A far-field reception geometry would make $\mathbf{u}_m^{(p)} \approx \mathbf{u}^{(p)}$ for all m . No such condition is assumed here, since viewing-direction diversity across apertures is what identifies the attitude.

Collecting the attitude-dependent factors of (10) gives the Backscatter-MIMO array manifold

$$[\Gamma(\mathbf{R}^{(p)})]_{m,k} \triangleq e^{-j\kappa_\lambda(\mathbf{u}_m^{(p)})^T \mathbf{R}^{(p)} \mathbf{b}_k}. \quad (11)$$

Stacking (6) over all apertures and chips then gives the raw received matrix $\mathbf{Y}^{(p)} \in \mathbb{C}^{M \times L}$,

$$\mathbf{Y}^{(p)} = \Psi^{(p)} (\mathbf{H}_{\text{rt}} \odot \Gamma(\mathbf{R}^{(p)})) \mathbf{C} + \mathbf{Z}^{c,(p)} + \mathbf{N}^{(p)}, \quad (12)$$

where $\Psi^{(p)} \triangleq \text{diag}(e^{-j\chi_1^{(p)}}, \dots, e^{-j\chi_M^{(p)}})$ carries the per-aperture common phase $\chi_m^{(p)} \triangleq \phi_m^{\text{trans},(p)} - \psi_m^{(p)}$, $[\mathbf{H}_{\text{rt}}]_{m,k} = h_{m,k}$ collects the calibrated gains, and $\mathbf{Z}^{c,(p)}$ and $\mathbf{N}^{(p)}$ stack the disturbance and noise samples. The nuisances sit in $\Psi^{(p)}$ and $\mathbf{H}_{\text{rt}}$, and the attitude sits in $\Gamma(\mathbf{R}^{(p)})$ alone. Of the channel parameters, the gains $\{h_{m,k}\}$ and the viewing directions $\{\mathbf{u}_m^{(p)}\}$ are known from calibration and tracking, the disturbance and noise are known through their second-order statistics $\mathbf{R}_\xi^{(m)}$ and σ^2 , and the common phases $\{\chi_m^{(p)}\}$, the clutter amplitudes $\{z_m^{s,(p)}\}$, and the disturbance realizations are unknown.

C. Problem Formulation

Wrapping makes attitude estimation a mixed integer-manifold problem. Angular-velocity estimation is linear once the attitudes are fixed. The receive side knows $\mathbf{a}_0$, $\{\mathbf{a}_m\}_{m=1}^M$, $\{\mathbf{b}_k\}_{k=1}^K$, $\mathbf{C}$, and λ from site and target calibration. Coarse geometric calibration or conventional tracking supplies $\{\mathbf{u}_m^{(p)}\}$. The second-order disturbance statistics are calibrated, for instance from empty-scene intervals. The instantaneous clutter, interference, and noise realizations are unknown, and so are the LO phases $\{\psi_m^{(p)}\}$.

The unknowns fall into three classes. The rotational states of interest are the attitudes $\{\mathbf{R}^{(p)} \in \mathcal{SO}(3)\}_{p=1}^P$ and the angular velocities $\{\boldsymbol{\omega}^{(p)} \in \mathbb{R}^3\}_{p=2}^P$, coupled by (5). The integer ambiguities $\mathbf{n} \in \mathbb{Z}^d$ arise because phases are observed

modulo 2π , with $d = M(K-1)$ the number of differential channels. The nuisances are the per-aperture common phases $\chi_m^{(p)}$ and the calibrated gains $h_{m,k}$.

Two structural facts shape the problem. The nuisances are removed by the construction of the observation rather than estimated, and once the integers are resolved this costs no attitude information. Only finitely many integer vectors can arise, each entry being bounded by $\lceil \kappa_\lambda \|\mathbf{u}_m^{(p)}\| \|\Delta \mathbf{b}_k\| / (2\pi) \rceil + 1$, because $\mathcal{SO}(3)$ is compact and the baselines are known, which is what makes the integer search tractable. Per-pulse attitude estimation therefore solves

$$\begin{aligned} & \min_{\mathbf{R}, \mathbf{n}} \|\mathbf{y}_R^{(p)} + 2\pi \mathbf{n} + \kappa_\lambda \mathbf{h}(\mathbf{R})\|_{\mathbf{W}_\varphi^{(p)}}^2 \\ & \text{s.t. } \mathbf{R}^T \mathbf{R} = \mathbf{I}_3, \quad \det(\mathbf{R}) = 1, \\ & \quad \mathbf{n} \in \mathbb{Z}^d, \end{aligned} \quad (13)$$

where $\mathbf{y}_R^{(p)} \in [-\pi, \pi)^d$ is the wrapped translation-free differential phase observation, $\mathbf{h}(\mathbf{R})$ has entries $(\mathbf{u}_m^{(p)})^T \mathbf{R} \Delta \mathbf{b}_k$ with the target-side baselines $\Delta \mathbf{b}_k$, both constructed in Section III-A, $\mathbf{W}_\varphi^{(p)}$ is the inverse of the propagated phase-noise covariance, and $\|\mathbf{x}\|_{\mathbf{W}}^2 \triangleq \mathbf{x}^T \mathbf{W} \mathbf{x}$. The first two constraints place $\mathbf{R}$ on $\mathcal{SO}(3)$ and the third carries the wrapping, and the minimizer is written $(\hat{\mathbf{R}}^{(p)}, \hat{\mathbf{n}})$.

Angular velocity enters linearly through the inter-pulse phase rates, so for adjacent pulses $p-1$ and p ,

$$\hat{\boldsymbol{\omega}}^{(p)} = \arg \min_{\boldsymbol{\omega} \in \mathbb{R}^3} \|\mathbf{y}_\omega^{(p)} - \mathbf{H}_\omega(\hat{\mathbf{R}}^{(p-1)}) \boldsymbol{\omega}\|_{\mathbf{W}_f^{(p)}}^2, \quad (14)$$

where $\mathbf{y}_\omega^{(p)}$ is the phase-rate observation and $\mathbf{H}_\omega(\cdot)$ its geometry matrix, both constructed in Section III-C, and $\mathbf{W}_f^{(p)}$ is the inverse of the phase-rate noise covariance.

III. FROM CODED ECHOES TO ATTITUDE AND ANGULAR-VELOCITY ESTIMATES

The raw echo becomes an estimate in three stages, one per subsection. One conjugate difference removes the per-aperture common phases. The resulting wrapped phases pose a mixed integer-manifold problem, solved by a grid search over $\mathcal{SO}(3)$ with integer least squares at every node under a deterministic covering condition. The estimator then runs at the pulse rate and reads the angular velocity off the phase rates.

A. Constructing the Wrapped Phase Observation

Four operations turn the raw echo matrix into a phase equation whose only unknowns are the attitude and a vector of integers. They are matched filtering, gain calibration, pairwise differencing, and phase extraction.

1) *Coded Echo and Matched filtering.* Separating the K coded antennas requires a row-orthogonal coding matrix,

$$\mathbf{C} \mathbf{C}^H = \mathbf{I}_{K'}, \quad (15)$$

which the phase-only DFT code $[\mathbf{C}]_{k,l} = e^{j2\pi k(l-1)/L}$ with $L \geq K+1$ realizes. We assign the K antennas to the DFT rows $2, \dots, K+1$ and leave the DC row unassigned. Static clutter is constant across the code block, so it lands entirely in the discarded DC output, $\frac{1}{L} \mathbf{z}^{s,(p)} \mathbf{1}_L^T \mathbf{C}^H = \mathbf{0}$, and all K antennas

survive the filter. Matched filtering $\hat{\mathbf{Y}}^{(p)} = \frac{1}{L} \mathbf{Y}^{(p)} \mathbf{C}^H$ applied to (12) then gives

$$\hat{\mathbf{Y}}^{(p)} = \boldsymbol{\Psi}^{(p)} (\mathbf{H}_{\text{rt}} \odot \boldsymbol{\Gamma}(\mathbf{R}^{(p)})) + \tilde{\boldsymbol{\Xi}}^{(p)} + \tilde{\mathbf{N}}^{(p)}, \quad (16)$$

where $\tilde{\boldsymbol{\Xi}}^{(p)} = \frac{1}{L} \boldsymbol{\Xi}^{(p)} \mathbf{C}^H$ and $\tilde{\mathbf{N}}^{(p)} = \frac{1}{L} \mathbf{N}^{(p)} \mathbf{C}^H$ are the post-filter interference and noise, with $[\boldsymbol{\Xi}^{(p)}]_{m,l} = \xi_m^{(p)}[l]$, $\tilde{n}_{m,k}^{(p)} \sim \mathcal{CN}(0, \sigma^2/L)$, and $\mathbb{E}[|\tilde{\xi}_{m,k}^{(p)}|^2] = L^{-2} \mathbf{c}_k^H \mathbf{R}_\xi^{(m)} \mathbf{c}_k$, where $\mathbf{c}_k \in \mathbb{C}^L$ is the transpose of the k -th row of $\mathbf{C}$. Entrywise,

$$\hat{y}_{m,k}^{(p)} = h_{m,k} e^{-j\chi_m^{(p)}} e^{-j\phi_{m,k}^{\text{rot},(p)}} + \tilde{\xi}_{m,k}^{(p)} + \tilde{n}_{m,k}^{(p)}. \quad (17)$$

2) *Gain calibration.* Dividing each decoded entry by its calibrated gain, $\bar{y}_{m,k}^{(p)} \triangleq \hat{y}_{m,k}^{(p)} / h_{m,k}$, leaves the signal term at unit modulus and rescales the perturbations.

3) *Pairwise differencing.* The translational phase and the LO phase are both column-common, so a single conjugate product removes them together. Pair each antenna $k \in \{2, \dots, K\}$ with a reference antenna $k' \in \{1, \dots, K\}$, and write $\Delta \mathbf{b}_k \triangleq \mathbf{b}_k - \mathbf{b}_{k'}$ for the resulting baseline. The differential product is

$$d_{m,k}^{(p)} \triangleq \bar{y}_{m,k}^{(p)} (\bar{y}_{m,k'}^{(p)})^* = e^{-j\kappa_\lambda \mathbf{u}_m^T \mathbf{R}^{(p)} \Delta \mathbf{b}_k} + \eta_{m,k}^{(p)}, \quad (18)$$

where $\eta_{m,k}^{(p)}$ collects the calibrated post-filter perturbations. The noiseless factor follows from $e^{-j\chi_m^{(p)}} e^{+j\chi_m^{(p)}} = 1$ together with $\phi_{m,k}^{\text{rot},(p)} - \phi_{m,k'}^{\text{rot},(p)} = \kappa_\lambda \mathbf{u}_m^T \mathbf{R}^{(p)} \Delta \mathbf{b}_k$.

What differencing discards is the common mode of each aperture. Once the integers are resolved, Proposition 2 shows that no attitude information leaves with it. Any spanning tree of the antenna set is an admissible pairing.

4) *Phase extraction and stacking.* Taking the angle of the differential product gives the wrapped phase $\tilde{\varphi}_{m,k}^{(p)} \triangleq \angle d_{m,k}^{(p)} \in [-\pi, \pi)$. Under small-noise linearization it obeys (18) up to an integer $n_{m,k}^*$,

$$\tilde{\varphi}_{m,k}^{(p)} + 2\pi n_{m,k}^* = -\kappa_\lambda \mathbf{u}_m^T \mathbf{R}^{(p)} \Delta \mathbf{b}_k + \epsilon_{m,k}^{(p)}, \quad (19)$$

where $\epsilon_{m,k}^{(p)}$ is the equivalent phase noise induced by operations 1)–3). Stacking over $m = 1, \dots, M$ and $k = 2, \dots, K$ gives

$$\tilde{\boldsymbol{\varphi}}^{(p)} + 2\pi \mathbf{n}^* = -\kappa_\lambda \mathbf{B} \text{vec}(\mathbf{R}^{(p)}) + \boldsymbol{\epsilon}^{(p)}, \quad (20)$$

where $\tilde{\boldsymbol{\varphi}}^{(p)} \in [-\pi, \pi)^{M(K-1)}$ and $\mathbf{n}^* \in \mathbb{Z}^{M(K-1)}$ stack the wrapped phases and the integers, $\boldsymbol{\epsilon}^{(p)}$ stacks the equivalent phase noise, and $\mathbf{B} \in \mathbb{R}^{M(K-1) \times 9}$ carries the two-sided geometry,

$$[\mathbf{B}]_{(m,k),:} = \Delta \mathbf{b}_k^T \otimes \mathbf{u}_m^T. \quad (21)$$

Each row is one Kronecker product of a target-side baseline with an aperture-side viewing direction. Translation, LO phase, and gain have all left (20), and only the attitude and the integers remain. Matched filtering, gain calibration, and differencing also color the noise, so the weight of (13) needs the covariance $\mathbf{C}_\varphi^{(p)} \triangleq \mathbb{E}[\boldsymbol{\epsilon}^{(p)} (\boldsymbol{\epsilon}^{(p)})^T]$. We propagate it from the chip level under proper-complex perturbations, small-noise phase linearization, resolved integer ambiguities, and receiver perturbations uncorrelated across apertures.

Proposition 1 (Equivalent phase-domain noise covariance). *Under these conditions, the equivalent noise $\boldsymbol{\epsilon}^{(p)}$ in (20) is zero-mean with covariance*

$$\mathbf{C}_\varphi^{(p)} = \text{blkdiag}(\mathbf{D}\mathbf{C}_\theta^{(1,p)}\mathbf{D}^T, \dots, \mathbf{D}\mathbf{C}_\theta^{(M,p)}\mathbf{D}^T), \quad (22)$$

where $\mathbf{D} \in \mathbb{R}^{(K-1) \times K}$ implements the pairwise differencing of the adopted scheme, with row $(\mathbf{e}_k - \mathbf{e}_{k'})^T$ for the pair (k, k') and $\mathbf{e}_k$ the k -th standard basis vector, and $\mathbf{C}_\theta^{(m,p)}$ is the covariance of the K -dimensional phase-noise vector across the decoded antenna columns at aperture m . Specifically,

$$\mathbf{C}_\theta^{(m,p)} = \frac{1}{2L^2} \text{Re} \{ \mathbf{S}_m^{(p)} \mathbf{H}_m^{-1} \mathbf{C}_{\text{cf}}^H \mathbf{R}_v^{(m)} \mathbf{C}_{\text{cf}} \mathbf{H}_m^{-H} \mathbf{S}_m^{(p)H} \}, \quad (23)$$

where $\mathbf{S}_m^{(p)} = \text{diag}(e^{j\phi_{m,1}^{\text{tot},(p)}}, \dots, e^{j\phi_{m,K}^{\text{tot},(p)}})$, $\phi_{m,k}^{\text{tot},(p)} = \phi_{m,k}^{\text{trans},(p)} + \phi_{m,k}^{\text{rot},(p)}$, $\mathbf{H}_m = \text{diag}(h_{m,1}, \dots, h_{m,K})$, $\mathbf{C}_{\text{cf}} = [\mathbf{c}_1, \dots, \mathbf{c}_K]$, and $\mathbf{R}_v^{(m)} = \mathbf{R}_\xi^{(m)} + \sigma^2 \mathbf{I}_L$.

Proof: The proof is provided in Appendix J.

Conditioned on the correct ambiguity branch and under the small-noise approximation, the wrapped and unwrapped phases share the same local second-order statistics, so $\mathbf{C}_\varphi^{(p)}$ also serves the integer search. The phase factor $\mathbf{S}_m^{(p)}$ is evaluated once, from calibrated phases, measured unwrapped phases, or a preliminary attitude estimate, and $\mathbf{C}_\varphi^{(p)}$ is then held fixed. It cancels when $\mathbf{R}_v^{(m)}$ is white and the codes are orthogonal.

B. Attitude Estimation and Integer Ambiguity Resolution

The wrapped vector $\tilde{\boldsymbol{\varphi}}^{(p)}$ of Section III-A is the observation $\mathbf{y}_R^{(p)}$, with $d = M(K-1)$ and $\mathbf{B} \text{vec}(\mathbf{R}) = \mathbf{h}(\mathbf{R})$, so problem (13) becomes

$$\begin{aligned} \min_{\mathbf{R}, \mathbf{n}} \quad & \|\tilde{\boldsymbol{\varphi}}^{(p)} + 2\pi \mathbf{n} + \kappa_\lambda \mathbf{B} \text{vec}(\mathbf{R})\|_{\mathbf{W}_\varphi^{(p)}}^2 \\ \text{s.t.} \quad & \mathbf{R}^T \mathbf{R} = \mathbf{I}_3, \quad \det(\mathbf{R}) = 1, \\ & \mathbf{n} \in \mathbb{Z}^d, \end{aligned} \quad (24)$$

where $\mathbf{W}_\varphi^{(p)} = (\mathbf{C}_\varphi^{(p)})^{-1}$. Algorithm 1 solves an integer least-squares problem at every grid node, searches the resulting three-dimensional cost over $\mathcal{SO}(3)$, and refines the best candidates on the manifold.

Integer least squares at every node. The minimization over $\mathbf{n}$ at fixed $\mathbf{R}_g$ is solved exactly rather than heuristically, and the resulting V_g is the concentrated, or profile, cost of separable estimation. The weight $\mathbf{W}_\varphi^{(p)}$ is block diagonal, so the minimization over $\mathbf{n}$ at fixed $\mathbf{R}_g$ splits into M integer least-squares problems of dimension $K-1$. Problem m carries the residual entries $\mathbf{r}_m$ of aperture m and the weight $\mathbf{W}_m^{(p)}$, the inverse of the m -th diagonal block of $\mathbf{C}_\varphi^{(p)}$ in (22). A diagonal $\mathbf{W}_m^{(p)}$ reduces the minimizer to the elementwise rounding $-\lfloor \mathbf{r}_m / (2\pi) \rfloor$. Under the white antenna-level noise of Proposition 1 without interference, the weighted cost equals the dispersion of the K unwrapped antenna phases about their mean. The minimizer is therefore one of the K cyclic cuts of the sorted wrapped antenna residuals, found exactly in $O(K \log K)$. V_g is then the exact partial minimum of (24) at $\mathbf{R}_g$, which leaves a three-dimensional outer search. At fixed $\mathbf{n}$,

Algorithm 1 Attitude determination with integer ambiguity resolution

Require: $\tilde{\varphi}^{(p)}$, $\mathbf{B}$, $\mathbf{W}_\varphi^{(p)}$, grid $\mathcal{G}_\delta$ from (26), candidate count N_c , damping μ , tolerance ε_{tol}
Ensure: $(\hat{\mathbf{R}}^{(p)}, \hat{\mathbf{n}})$

- 1: $\mathbf{B}_\mathbf{W}^\dagger \leftarrow -\kappa_\lambda^{-1}(\mathbf{B}^T \mathbf{W}_\varphi^{(p)} \mathbf{B})^\dagger \mathbf{B}^T \mathbf{W}_\varphi^{(p)}$
- 2: **for** $\mathbf{R}_g \in \mathcal{G}_\delta$ **do**
- 3: $\mathbf{r} \leftarrow \tilde{\varphi}^{(p)} + \kappa_\lambda \mathbf{B} \text{vec}(\mathbf{R}_g)$
- 4: $\hat{\mathbf{n}}_m \leftarrow \arg \min_{\mathbf{n} \in \mathbb{Z}^{K-1}} \|\mathbf{r}_m + 2\pi \mathbf{n}\|_{\mathbf{W}_m^{(p)}}^2, \quad m = 1, \dots, M$
- 5: $V_g \leftarrow \sum_{m=1}^M \|\mathbf{r}_m + 2\pi \hat{\mathbf{n}}_m\|_{\mathbf{W}_m^{(p)}}^2$
- 6: **end for**
- 7: keep the N_c candidates $(\mathbf{R}_g, \hat{\mathbf{n}})$ with the smallest V_g
- 8: **for** each kept candidate **do**
- 9: $\text{vec}(\hat{\mathbf{R}}) \leftarrow \mathbf{B}_\mathbf{W}^\dagger (\tilde{\varphi}^{(p)} + 2\pi \hat{\mathbf{n}})$
- 10: $\mathbf{R} \leftarrow \mathbf{U} \text{diag}(1, 1, \det(\mathbf{UV}^T)) \mathbf{V}^T$ with $\tilde{\mathbf{R}} = \mathbf{U} \mathbf{\Lambda} \mathbf{V}^T$
- 11: **repeat**
- 12: $\mathbf{e}_{m,k} \leftarrow \tilde{\varphi}_{m,k}^{(p)} + 2\pi \hat{\mathbf{n}}_{m,k} + \kappa_\lambda \mathbf{u}_m^T \mathbf{R} \Delta \mathbf{b}_k$, stacked in $\mathbf{e}$
- 13: $\mathbf{J}_{m,k} \leftarrow \kappa_\lambda (\mathbf{R} \Delta \mathbf{b}_k \times \mathbf{u}_m)^T$, stacked in $\mathbf{J}$
- 14: $\delta \leftarrow -(\mathbf{J}^T \mathbf{W}_\varphi^{(p)} \mathbf{J} + \mu \mathbf{I}_3)^{-1} \mathbf{J}^T \mathbf{W}_\varphi^{(p)} \mathbf{e}, \quad \mathbf{R} \leftarrow \text{Exp}(\delta) \mathbf{R}$
- 15: **until** $\|\delta\| \leq \varepsilon_{\text{tol}}$
- 16: $q_g \leftarrow \|\mathbf{e}\|_{\mathbf{W}_\varphi^{(p)}}^2$
- 17: **end for**
- 18: **return** the kept pair $(\hat{\mathbf{R}}^{(p)}, \hat{\mathbf{n}})$ attaining $\min_g q_g$

conversely, (24) is a linear least-squares problem on $\mathcal{SO}(3)$. The integer assignment is the main source of combinatorial nonconvexity; for fixed $\mathbf{n}$, the remaining problem is a smooth constrained least-squares problem on $\mathcal{SO}(3)$.

Grid resolution. The resolution of the grid need not be tuned. Let $\mathcal{G}_\delta \subset \mathcal{SO}(3)$ have covering radius δ in the geodesic metric, let $\mathbf{R}_g$ be its point closest to the true attitude $\mathbf{R}^{(p)}$, and let $\mathbf{r}(\mathbf{R}_g)$ be the residual computed there. The model (20) gives

$$\mathbf{r}(\mathbf{R}_g) + 2\pi \mathbf{n}^* = \epsilon^{(p)} + \kappa_\lambda \mathbf{B} \text{vec}(\mathbf{R}_g - \mathbf{R}^{(p)}), \quad (25)$$

whose entry at the pair (m, k) departs from the noise $\epsilon_{m,k}^{(p)}$ by at most $\kappa_\lambda \|\mathbf{u}_m\| \|\Delta \mathbf{b}_k\| \|\mathbf{R}_g - \mathbf{R}^{(p)}\|_2$ in magnitude. Two rotations at geodesic distance $\delta_g \leq \delta$ satisfy $\|\mathbf{R}_g - \mathbf{R}^{(p)}\|_2 = 2 \sin(\delta_g/2) \leq \delta$. The grid therefore suffices whenever

$$\delta \leq \frac{\pi - \gamma}{\kappa_\lambda \max_m \|\mathbf{u}_m\| \max_k \|\Delta \mathbf{b}_k\|} \quad (26)$$

holds for some margin $\gamma \in (0, \pi)$. Every entry of $\mathbf{r}(\mathbf{R}_g) + 2\pi \mathbf{n}^*$ then stays below π in magnitude as long as $|\epsilon_i^{(p)}| \leq \gamma$ for all i , and the per-aperture rounding at $\mathbf{R}_g$ returns $\mathbf{n}^*$ exactly. A union bound over the d Gaussian tails places the probability of that event no lower than $1 - 2 \sum_{i=1}^d Q(\gamma/\sigma_i)$, with $\sigma_i^2 = [\mathbf{C}_\varphi^{(p)}]_{ii}$ and $Q(\cdot)$ the Gaussian tail function.

Scope of the guarantee. Condition (26) secures the integers at the grid point nearest the truth. It does not make the concentrated cost globally optimal. A wrong candidate may still attain a smaller refined cost, and the probability that none does is the ambiguity success rate $P_s \triangleq \Pr\{\hat{\mathbf{n}} = \mathbf{n}^*\}$. It degrades as σ_φ or the projected aperture $\max_k \|\Delta \mathbf{b}_k\|/\lambda$ grows, and improves with more antennas and better conditioning of $\mathbf{B}$.

The refined estimator. With $\hat{\mathbf{n}}$ fixed at a kept candidate, (24) reduces to the constrained weighted least-squares problem

$$\hat{\mathbf{R}}_{\text{WLS}}^{(p)} = \arg \min_{\mathbf{R} \in \mathcal{SO}(3)} \left\| \tilde{\varphi}^{(p)} + 2\pi \hat{\mathbf{n}} + \kappa_\lambda \mathbf{B} \text{vec}(\mathbf{R}) \right\|_{\mathbf{W}_\varphi^{(p)}}^2, \quad (27)$$

which is the maximum-likelihood estimator of $\mathbf{R}^{(p)}$ under the Gaussian phase model of Proposition 1 with the covariance held fixed. Whether its closed-form relaxation identifies the attitude depends on $\mathbf{B}$ alone. When $\text{rank}(\mathbf{B}) = 9$, the relaxed estimate is linearly identifiable. When $\text{rank}(\mathbf{B}) < 9$, the Procrustes projection, computed from the singular value decomposition of the relaxed estimate, serves as an efficient initializer for the constrained $\mathcal{SO}(3)$ estimator. We refer to this covariance-aware WLS–Procrustes initialization followed by manifold refinement as WLS+GN.

C. Recursive Tracking and Angular Velocity

After initialization the estimator runs recursively at the pulse rate, and the integer search is never repeated. The attitude is predicted through (5) as $\mathbf{R}_{\text{pred}} = \text{Exp}(T_{\text{PRI}} \hat{\omega}^{(p-1)}) \hat{\mathbf{R}}^{(p-1)}$, and the integers are propagated by the rounding $-\lfloor \mathbf{r}(\mathbf{R}_{\text{pred}})/(2\pi) \rfloor$. The prediction warm-starts the Gauss–Newton iterations of Algorithm 1, at $O(d)$ cost per iteration. Propagation is exact as long as every entry of the inter-pulse phase change stays below π , which the slow-rotation bound $\kappa_\lambda \|\omega\| T_{\text{PRI}} \max_m \|\mathbf{u}_m\| \max_k \|\Delta \mathbf{b}_k\| \ll \pi$ secures. The fix is validated at every pulse. Under correct integers, and for $\text{rank}(\mathbf{G}_R) = 3$, the refined cost q is asymptotically χ^2 with $d - 3$ degrees of freedom. A value $q > \chi_{d-3, 1-\alpha}^2$ flags a wrong fix and triggers re-initialization.

Angular velocity follows from the finite difference of two consecutive differential phase vectors, $\mathbf{f}^{(p)} \triangleq (\tilde{\varphi}^{(p)} - \tilde{\varphi}^{(p-1)})/T_{\text{PRI}}$ for $p = 2, \dots, P$. The phase rates are linear in the angular velocity under three conditions: the kinematic update (5), the slow-rotation condition $\|\omega^{(p)}\| T_{\text{PRI}} \ll 1$, and effective viewing directions that stay constant or are compensated over one PRI. Stacking the rates gives

$$\mathbf{f}^{(p)} = \mathbf{H}_\omega \left(\mathbf{R}^{(p-1)} \right) \omega^{(p)} + \nu^{(p)}, \quad (28)$$

where $\nu^{(p)} = (\epsilon^{(p)} - \epsilon^{(p-1)})/T_{\text{PRI}}$, and $\mathbf{H}_\omega(\mathbf{R}^{(p-1)}) \in \mathbb{R}^{M(K-1) \times 3}$ has row $-\kappa_\lambda (\mathbf{R}^{(p-1)} \Delta \mathbf{b}_k \times \mathbf{u}_m)^T$ for the pair (m, k) . The differencing of Section III-A also removes the common range-rate Doppler. Because $\chi_m^{(p)}$ is eliminated afresh at every pulse, the phase rate is likewise immune to slow LO drift across pulses. The vector $\mathbf{f}^{(p)}$ is the inter-pulse observation $\mathbf{y}_\omega^{(p)}$ of (14). It is built from $\{\tilde{\varphi}^{(p)}\}$ and carries no information beyond it.

Adjacent-pulse phase noises are independent under Proposition 1, so the phase-rate covariance is

$$\mathbf{C}_f^{(p)} = (\mathbf{C}_\varphi^{(p)} + \mathbf{C}_\varphi^{(p-1)})/T_{\text{PRI}}^2, \quad (29)$$

which becomes $2\mathbf{C}_\varphi/T_{\text{PRI}}^2$ under stationarity. The angular velocity over the interval from pulse $p - 1$ to pulse p then follows causally from

$$\hat{\omega}^{(p)} = \arg \min_{\omega \in \mathbb{R}^3} \left\| \mathbf{f}^{(p)} - \mathbf{H}_\omega \left(\hat{\mathbf{R}}^{(p-1)} \right) \omega \right\|_{\mathbf{W}_f^{(p)}}^2, \quad (30)$$

with $\mathbf{W}_f^{(p)} = (\mathbf{C}_f^{(p)})^{-1}$. Problem (30) is linear in ω and solved in closed form. The Moore–Penrose inverse returns the minimum-norm estimate in the observable subspace for ill-conditioned geometries. The estimator is a plug-in one, since

$\mathbf{H}_\omega$ takes the previous attitude estimate, so the slow-rotation approximation and the attitude error enter its accuracy.

IV. OBSERVABILITY AND INTRINSIC PERFORMANCE LIMITS

Two results govern rotational sensing, and both are set by the two sides of the geometry. Identifiability is a dimension count. The attitude is recoverable exactly when the two sides together span five dimensions, and the same count makes the Fisher information nonsingular. Accuracy is a product of an aperture-side factor and a target-side factor, and a one-dimensional span on either side loses an axis that the other side cannot restore. Below the threshold, the ambiguities form a short catalog; above it, every bound holds verbatim for phase-incoherent apertures.

A. Identifiability from the Two-Sided Dimensions

Identifiability splits into two questions. The model (20) is linear in $\text{vec}(\mathbf{R}^{(p)})$, and the relaxed step of Algorithm 1 treats that vector as a free Euclidean unknown. The first question is when this linear map is injective. The second is what the constraint $\mathbf{R} \in \mathcal{SO}(3)$ recovers beyond that point. Both answers enter only through the spanned subspaces

$$(U, V) \triangleq (\text{span}\{\mathbf{u}_m\}_{m=1}^M, \text{span}\{\Delta\mathbf{b}_k\}_{k=2}^K), \quad (31)$$

with the dimension pair $(d_u, d_b) \triangleq (\dim U, \dim V)$. The pulse index of $\mathbf{u}_m^{(p)}$ is suppressed throughout this section.

Lemma 1 (Unconstrained rank). *The matrix $\mathbf{B}$ of (20) has $\text{rank}(\mathbf{B}) = d_u d_b$, so $\text{vec}(\mathbf{R}^{(p)})$ is uniquely recoverable if and only if $d_u = d_b = 3$.*

Proof: By (21) the rows of $\mathbf{B}$ span $V \otimes U$, of dimension $d_b d_u$, and a free vector in $\mathbb{R}^9$ is recoverable if and only if $\text{rank}(\mathbf{B}) = 9$. ■

Linear recovery therefore needs $M \geq 3$ and $K \geq 4$. The constraint $\mathbf{R} \in \mathcal{SO}(3)$ removes part of that requirement. Observability concerns the noiseless observation map

$$\mathcal{T}(\mathbf{R}) \triangleq \{-\kappa_\lambda \mathbf{u}_m^T \mathbf{R} \Delta \mathbf{b}_k\}_{m=1, k=2}^{M, K}, \quad (32)$$

and the attitude is globally identifiable when $\mathcal{T}$ is injective on $\mathcal{SO}(3)$. Write $\mathcal{A}(\mathbf{R}_0) \triangleq \mathcal{T}^{-1}(\mathcal{T}(\mathbf{R}_0))$, the preimage of $\mathcal{T}(\mathbf{R}_0)$ under $\mathcal{T}$, for the attitude ambiguity set at $\mathbf{R}_0$, collecting the attitudes that no noiseless observation separates from $\mathbf{R}_0$. Injectivity says, by definition, that every such set is the singleton $\{\mathbf{R}_0\}$. Identifiability refers to this unwrapped map, that is, to the attitude given the integer vector $\mathbf{n}^*$. The many-to-one wrapping of (20) is the integer ambiguity of Section III-B, resolved statistically with success rate P_s .

Accuracy needs the first-order picture as well. Under the left perturbation $\mathbf{R}(\boldsymbol{\vartheta}) = \text{Exp}(\boldsymbol{\vartheta})\mathbf{R}_0$, the Jacobian of the differential phase $-\kappa_\lambda \mathbf{u}_m^T \mathbf{R} \Delta \mathbf{b}_k$ with respect to $\boldsymbol{\vartheta}$ is $-\kappa_\lambda (\mathbf{R}_0 \Delta \mathbf{b}_k \times \mathbf{u}_m)^T$. Stacking these rows in the order of (20) gives the local sensitivity matrix $\mathbf{G}_R(\mathbf{R}_0) \in \mathbb{R}^{M(K-1) \times 3}$,

$$[\mathbf{G}_R(\mathbf{R}_0)]_{(m,k),:} \triangleq (\mathbf{R}_0 \Delta \mathbf{b}_k \times \mathbf{u}_m)^T. \quad (33)$$

With $\mathbf{C}_\varphi^{(p)}$ fixed at the adopted phase-alignment point, the phase-domain FIM for $\boldsymbol{\vartheta}$ is

$$\mathbf{J}_R^{(p)}(\mathbf{R}_0) = \kappa_\lambda^2 \mathbf{G}_R^T(\mathbf{R}_0) (\mathbf{C}_\varphi^{(p)})^{-1} \mathbf{G}_R(\mathbf{R}_0), \quad (34)$$

since the stacked Jacobian of (20) is $-\kappa_\lambda \mathbf{G}_R(\mathbf{R}_0)$ and the phase noise is Gaussian. The dependence of $\mathbf{C}_\varphi^{(p)}$ on $\mathbf{R}$ through $\mathbf{S}_m^{(p)}$ contributes a term of order one to the FIM, against the order σ_φ^{-2} of (34), and vanishes in the white-noise regime of Section IV-C. It is omitted throughout.

The two recovery diagnostics admit one description each, and both read off the pair (U, V) .

Lemma 2 (Ambiguity set and null space). *For $\mathbf{R}, \mathbf{R}', \mathbf{R}_0 \in \mathcal{SO}(3)$ and positive definite $\mathbf{C}_\varphi^{(p)}$, the following hold.*

- (a) $\mathcal{T}(\mathbf{R}) = \mathcal{T}(\mathbf{R}')$ if and only if $(\mathbf{R} - \mathbf{R}')V \subseteq U^\perp$.
- (b) $\ker \mathbf{J}_R^{(p)}(\mathbf{R}_0) = \ker \mathbf{G}_R(\mathbf{R}_0) = \mathcal{S}(\mathbf{R}_0)^\perp$, where

$$\mathcal{S}(\mathbf{R}_0) \triangleq \text{span}\{\mathbf{a} \times \mathbf{u} : \mathbf{a} \in \mathbf{R}_0 V, \mathbf{u} \in U\}. \quad (35)$$

Proof: The proof is provided in Appendix A.

Statement (a) is the global diagnostic. It turns membership of $\mathcal{A}(\mathbf{R}_0)$ into an inclusion between subspaces. Statement (b) is the local diagnostic. It collects the locally unobservable rotation axes in the orthogonal complement of a cross-product span. Both statements are symmetric in the two sides. Criterion (a) is an orthogonality between $(\mathbf{R} - \mathbf{R}')V$ and U , and criterion (b) is unchanged when $\mathbf{R}_0 V$ and U swap roles. Every two-sided statement below inherits this symmetry. Counting dimensions in Lemma 2 collapses both diagnostics into one integer condition. Let $\mathbf{C}_\varphi^{(p)}$ be positive definite and held fixed.

Theorem 1 (Observability threshold). *Under this condition, the following three statements are equivalent.*

- (i) $d_u + d_b \geq 5$.
- (ii) The map $\mathcal{T}$ of (32) is injective on $\mathcal{SO}(3)$.
- (iii) The matrix $\mathbf{J}_R^{(p)}(\mathbf{R}_0)$ of (34) is nonsingular at every $\mathbf{R}_0 \in \mathcal{SO}(3)$.

Proof: The proof is provided in Appendix B.

Comparing Lemma 1 with statement (i) shows what the manifold buys. Unconstrained recovery needs the product of the two dimensions to reach nine, which forces both sides to span $\mathbb{R}^3$. The rotation constraint replaces that product by a sum, so one side may drop to two dimensions once the other spans $\mathbb{R}^3$. The condition is symmetric, and neither side is privileged. Of the twelve pairs $(d_u, d_b) \in \{1, 2, 3\} \times \{0, 1, 2, 3\}$, the product condition admits (3, 3) alone and the sum condition admits (2, 3), (3, 2), and (3, 3).

The equivalence of (ii) and (iii) is the operative part. A geometry that identifies the attitude at all identifies it with finite accuracy along every axis, so one integer threshold decides both. Noise enters statement (iii) only through the positive definite $\mathbf{C}_\varphi^{(p)}$ of Proposition 1, so the threshold does not move with the noise level.

The identifiability of the angular velocity follows from that of the attitude. The kinematics $\dot{\mathbf{R}}(t) = [\omega(t)]_\times \mathbf{R}(t)$ make it the time derivative of the tangent-space perturbation. The

angular-velocity sensing matrix of (28) is therefore the same object as (33),

$$\mathbf{H}_\omega(\mathbf{R}) = -\kappa_\lambda \mathbf{G}_R(\mathbf{R}), \quad (36)$$

so the two share one null space. Every statement above applies to the angular velocity unchanged, and the Moore–Penrose solution of (30) returns only its observable component.

B. Degenerate Geometries and Their Ambiguities

Below the threshold, the two diagnostics of Lemma 2 no longer agree, so the catalog lists both. The ambiguity set names the attitudes that the observation cannot separate, and the null space names the rotation axes that it cannot sense. Let $\mathbf{C}_\varphi^{(p)}$ be positive definite and let $d_u + d_b \leq 4$.

Corollary 1 (Degeneracy catalog). *Under these conditions, the attitude ambiguity set and the sensitivity null space are*

(d_u, d_b)	$d_u + d_b$	$\mathcal{A}(\mathbf{R}_0)$	$\ker \mathbf{G}_R(\mathbf{R}_0)$
$(\cdot, 0)$	≤ 3	$= \mathcal{SO}(3)$	$= \mathbb{R}^3$
$(1, \cdot)$	≤ 4	$\supseteq \{\text{Exp}(\theta \hat{\mathbf{u}}) \mathbf{R}_0\}_\theta$	$\supseteq \text{span}\{\hat{\mathbf{u}}\}$
$(\cdot, 1)$	≤ 4	$\supseteq \{\mathbf{R}_0 \text{Exp}(\theta \Delta \hat{\mathbf{b}})\}_\theta$	$\supseteq \text{span}\{\mathbf{R}_0 \Delta \hat{\mathbf{b}}\}$
$(2, 2)$	4	two attitudes, (51)	$= \{\mathbf{0}\}$
$(2, 2)^\dagger$	4	$= \{\mathbf{R}_0\}$	$= U$

† at the attitudes with $\mathbf{R}_0 V = U$.

and no other case occurs, where $\hat{\mathbf{u}}$ and $\Delta \hat{\mathbf{b}}$ generate the one-dimensional spans, with equality at $d_u + d_b = 4$.

Proof: The proof is provided in Appendix C.

The two one-parameter rows are duals. Spinning the body about a single viewing axis moves no phase, and rotating a collinear array about its own axis moves no antenna. The $(1, \cdot)$ row is classical. A single vector observation leaves the rotation about its own direction undetermined. The catalog adds what the classical statement lacks. No target-side change repairs this row, since its entries are independent of K and of the layout.

The two $(2, 2)$ rows are the point where the two diagnostics separate. At almost every attitude a discrete partner exists while the FIM stays nonsingular. On the one-dimensional set $\mathbf{R}_0 V = U$ the attitude is unique while the FIM is singular. That set is exactly where the partner merges with the truth. A $d_u = d_b = 2$ layout is therefore doubly ill conditioned near $\mathbf{R}_0 V \approx U$, carrying a nearly singular FIM together with a nearly coincident false solution.

The analysis depends on the reception geometry only through the set $\{\mathbf{u}_m\}$. The elements of one large receive array, the antennas of physically separated reception points, and the views accumulated over time by target motion all enter identically. Any fixed illumination whose phase is locally linear over the target aperture also fits, with $\hat{\mathbf{r}}_0$ read as the normalized illumination phase gradient at the CoM.

C. Intrinsic Performance Limits

The accuracy limit departs from point-target bounds in two ways. It is a product of an aperture-side factor and a target-side factor, and it depends on the attitude being estimated.

With the attitude error $\tilde{\boldsymbol{\vartheta}}^{(p)} \triangleq \text{Log}(\hat{\mathbf{R}}^{(p)}(\mathbf{R}^{(p)})^T)$, the tangent-space CRB is $\mathbf{P}_R^{(p)} \triangleq (\mathbf{J}_R^{(p)})^{-1}$. For locally unbiased

estimators, $\text{tr}(\mathbf{P}_R^{(p)})$ lower-bounds the mean squared geodesic error within the principal branch up to curvature terms of order $\|\mathbf{P}_R^{(p)}\|^2$ [27], and is the intrinsic bound at high SNR. The left-trivialized perturbation of (2) carries no coordinate singularity, so every singularity below is geometric.

Take the thermal-noise-limited, uniform-gain regime with the star pairing about one reference antenna. Clutter and interference are absent, $|h_{m,k}| = \rho$, and $\sigma_\varphi^2 \triangleq \sigma^2/(2L\rho^2)$, so Proposition 1 gives $\mathbf{C}_\varphi^{(m)} = \sigma_\varphi^2(\mathbf{I}_{K-1} + \mathbf{1}_{K-1}\mathbf{1}_{K-1}^T)$. The target side enters through the centered spread $\Sigma_{\text{eff}} \triangleq \Sigma - \boldsymbol{\mu}\boldsymbol{\mu}^T/K$, where $\Sigma \triangleq \sum_{k=2}^K \Delta \mathbf{b}_k \Delta \mathbf{b}_k^T$ has $\text{rank}(\Sigma) = d_b$ and $\boldsymbol{\mu} \triangleq \sum_{k=2}^K \Delta \mathbf{b}_k$.

Theorem 2 (Intrinsic attitude FIM). *Under these conditions, the per-pulse tangent-space FIM is*

$$\mathbf{J}_R(\mathbf{R}_0) = \frac{\kappa_\lambda^2}{\sigma_\varphi^2} \sum_{m=1}^M [\mathbf{u}_m]_\times \mathbf{R}_0 \Sigma_{\text{eff}} \mathbf{R}_0^T [\mathbf{u}_m]_\times^T. \quad (37)$$

Equivalently, $\text{vec}(\mathbf{J}_R) = (\kappa_\lambda^2/\sigma_\varphi^2) \boldsymbol{\Psi} \text{vec}(\mathbf{R}_0 \Sigma_{\text{eff}} \mathbf{R}_0^T)$, where $\boldsymbol{\Psi} \triangleq \sum_{m=1}^M [\mathbf{u}_m]_\times \otimes [\mathbf{u}_m]_\times \in \mathbb{R}^{9 \times 9}$ depends on the aperture side alone.

Proof: The proof is provided in Appendix D.

The rotated spread $\mathbf{R}_0 \Sigma_{\text{eff}} \mathbf{R}_0^T$ sets how strongly a rotation modulates the differential phases, and $\boldsymbol{\Psi}$ sets which rotation axes that modulation reaches. A rotation axis is lost exactly when it lies in the plane of every pair $(\mathbf{u}_m, \mathbf{R}_0 \Delta \mathbf{b}_k)$, which is Lemma 2(b) restated. A one-dimensional span on either side therefore loses its own axis whatever the other side does, while a planar spread loses none once the views are diverse. The conjugation by $\mathbf{R}_0$ makes the bound depend on the attitude being estimated. For anisotropic layouts it varies over $\mathcal{SO}(3)$; at $d_u = d_b = 2$ it becomes singular on the one-dimensional set of attitudes with $\mathbf{R}_0 V = U$ listed in Corollary 1. Isotropy of Σ_{eff} is exactly the condition that removes the dependence.

For a planar layout with rotated normal $\mathbf{n} \triangleq \mathbf{R}_0 \mathbf{e}_z$, write each viewing direction as $\mathbf{u}_m = \|\mathbf{u}_m\|(\cos \theta_m \mathbf{n} + \sin \theta_m \mathbf{s}_m)$. Here $\mathbf{s}_m \in \mathbf{n}^\perp$ is a unit vector and θ_m the off-broadside angle. Each row of (33) then splits into a spin-about- $\mathbf{n}$ part scaling with $\sin \theta_m$ and an in-plane part scaling with $\cos \theta_m$. Broadside views therefore leave the normal spin unobservable, grazing views weaken the in-plane axes, and a planar layout needs off-broadside views with diverse in-plane directions.

Call the centered spread isotropic when $\Sigma_{\text{eff}} = \text{tr}(\Sigma_{\text{eff}})\mathbf{I}_3/3$. Isotropy forces $d_b = 3$ and makes $\mathbf{R}_0 \Sigma_{\text{eff}} \mathbf{R}_0^T = \Sigma_{\text{eff}}$ at every attitude. The identity $\boldsymbol{\Psi} \text{vec}(\mathbf{I}_3) = \text{vec}(\boldsymbol{\Phi})$ then collapses Theorem 2 onto the aperture-side diversity matrix

$$\boldsymbol{\Phi} \triangleq \sum_{m=1}^M (\|\mathbf{u}_m\|^2 \mathbf{I}_3 - \mathbf{u}_m \mathbf{u}_m^T), \quad (38)$$

and the CRB reduces to

$$\mathbf{P}_R = \frac{3\sigma_\varphi^2}{\kappa_\lambda^2 \text{tr}(\Sigma_{\text{eff}})} \boldsymbol{\Phi}^{-1}. \quad (39)$$

$\boldsymbol{\Phi}$ has rank 3 if and only if $d_u \geq 2$, which is statement (i) of Theorem 1 once more. At $d_u = 1$ the bound holds on the observable subspace and is unbounded along $\hat{\mathbf{u}}$.

Discarding the common phases χ_m costs no attitude information. Let the unwrapped phases of aperture m be $\theta_m = \chi_m \mathbf{1}_K + \mathbf{g}_m(\boldsymbol{\vartheta}) + \varepsilon_m$, where χ_m is unknown, $[\mathbf{g}_m(\boldsymbol{\vartheta})]_k = -\kappa \lambda \mathbf{u}_m^T \text{Exp}(\boldsymbol{\vartheta}) \mathbf{R}_0 \mathbf{b}_k$, and $\varepsilon_m \sim \mathcal{N}(\mathbf{0}, \mathbf{C}_\theta^{(m)})$, $\mathbf{C}_\theta^{(m)} \succ \mathbf{0}$, independent across apertures. Let $\mathbf{D} \in \mathbb{R}^{(K-1) \times K}$ be any pairing matrix with rank $K - 1$ and $\mathbf{D} \mathbf{1}_K = \mathbf{0}$.

Proposition 2 (Attitude information in phase differences). *Under these conditions, the attitude FIM of the differenced phases $\mathbf{D}\boldsymbol{\theta}_m$ equals the Schur complement of the joint FIM of $(\boldsymbol{\vartheta}, \chi_1, \dots, \chi_M)$ with respect to the common phases. Hence a network without carrier coherence attains the attitude CRB of a coherent network that estimates $\{\chi_m\}_{m=1}^M$ jointly with the attitude. If moreover $\sum_{k=1}^K \mathbf{b}_k = \mathbf{0}$ and $\mathbf{C}_\theta^{(m)} = \sigma_\varphi^2 \mathbf{I}_K$, the attitude and the common phases are information orthogonal, and knowing χ_m would not tighten the bound.*

Proof: The proof is provided in Appendix E.

The common phase of an aperture is the two-way phase of the array centroid, which no rotation about the centroid moves. Thus, no inter-aperture carrier-phase coherence is required beyond common pulse timing and the calibration data.

Angular velocity is bounded from two consecutive pulses, and by (36) it inherits the same factorization. Collect $\boldsymbol{\xi}^{(p)} \triangleq [(\boldsymbol{\vartheta}^{(p-1)})^T, (\delta\boldsymbol{\omega}^{(p)})^T]^T$ and write $T_p \triangleq T_{\text{PRI}}$. Let the phase noise be temporally white and stationary, with the two pulse-wise noise vectors independent with common covariance $\mathbf{C}_\varphi$. Let the rotation be slow enough that $\mathbf{J}_R^{(p)} \approx \mathbf{J}_R^{(p-1)} \triangleq \mathbf{J}_R$, with $\mathbf{J}_R$ nonsingular by Theorem 1, and that $\boldsymbol{\vartheta}^{(p-1)}$ and $T_p \delta\boldsymbol{\omega}^{(p)}$ enter the phase identically.

Corollary 2 (Two-pulse tangent-space CRB). *Under these conditions, the joint FIM for $\boldsymbol{\xi}^{(p)}$ is*

$$\mathbf{J}^{(p)} = \begin{bmatrix} 2\mathbf{J}_R & T_p \mathbf{J}_R \\ T_p \mathbf{J}_R & T_p^2 \mathbf{J}_R \end{bmatrix}. \quad (40)$$

Eliminating $\delta\boldsymbol{\omega}^{(p)}$ leaves the attitude bound unchanged at $\mathbf{P}_{\boldsymbol{\vartheta}, \text{eff}} = \mathbf{P}_R$, and eliminating $\boldsymbol{\vartheta}^{(p-1)}$ gives $\mathbf{P}_\omega = 2\mathbf{P}_R/T_p^2$.

Proof: The proof is provided in Appendix F.

Since (27) is the maximum-likelihood criterion for the phase model of Proposition 1 with fixed covariance, its high-SNR error approaches these bounds once the integers are resolved.

V. GEOMETRY DESIGN LAWS

The factorization of Theorem 2 reduces geometry design to three choices: the target-side layout, the differencing topology, and the aperture-side placement. The baseline budget is shared between the information strength and the integer search, and only the topology decides which baselines the search retains. The bistatic angle trades viewing diversity against phase sensitivity, with its optimum strictly inside $(0, \pi)$.

A. Target-Side Layout

A layout enters the bound only through $\boldsymbol{\Sigma}_{\text{eff}}$: its trace sets the accuracy level, and its eigenvalue ratios set how evenly it covers attitudes. Let $\mathbf{c} \triangleq \frac{1}{K} \sum_{k=1}^K \mathbf{b}_k$ be the centroid.

Proposition 3 (Target-side spread law). *The centered spread $\boldsymbol{\Sigma}_{\text{eff}}$ of Section IV-C equals the centroid spread*

$$\boldsymbol{\Sigma}_{\text{eff}} = \sum_{k=1}^K (\mathbf{b}_k - \mathbf{c})(\mathbf{b}_k - \mathbf{c})^T, \quad (41)$$

independent of which antenna serves as the reference, with $\text{rank}(\boldsymbol{\Sigma}_{\text{eff}}) = d_b$. If every antenna lies within distance R of $\mathbf{c}$, then $\text{tr}(\boldsymbol{\Sigma}_{\text{eff}}) \leq KR^2$, with equality when all antennas sit on that sphere. At $K = 4$, $\boldsymbol{\Sigma}_{\text{eff}}$ is isotropic if and only if the antennas form a regular tetrahedron.

Proof: The proof is provided in Appendix G.

The design object is therefore the antenna point set, and the reference antenna is free. The trace grows quadratically with the mounting radius but only linearly with the antenna count, so a wide sparse array outperforms a compact dense one of equal hardware budget. Isotropy is the only spectrum invariant under the conjugation $\mathbf{R}_0 \boldsymbol{\Sigma}_{\text{eff}} \mathbf{R}_0^T$ of Theorem 2, so the bound is attitude independent exactly when $\boldsymbol{\Sigma}_{\text{eff}}$ is isotropic. The deviation is quantified by the volumetricity

$$\eta_\Sigma \triangleq \frac{3\lambda_{\min}(\boldsymbol{\Sigma}_{\text{eff}})}{\text{tr}(\boldsymbol{\Sigma}_{\text{eff}})} \in [0, 1], \quad (42)$$

which equals one at isotropy and zero whenever $d_b < 3$. The regular tetrahedron reaches full rank and isotropy together at $K = 4$, and further antennas enter the bound only through the trace. The angular-velocity bound inherits the same design amplified by $2/T_{\text{PRI}}^2$, so lengthening T_{PRI} helps only while the slow-rotation approximation holds.

B. Optimal Differencing Topology

Of the $K(K-1)/2$ pairwise baselines, differencing keeps $K-1$, and a valid pairing is a spanning tree, since a disconnected pairing loses rank and a cycle is redundant. Accuracy is indifferent to the pairing; the integer search is not.

Proposition 4 (Differencing topology). *The following hold over the spanning trees of the antenna set.*

- (a) *The unwrapped differenced observations of any two trees are related by an invertible linear map. Full-covariance weighting leaves the criterion (27) and the attitude FIM unchanged, and the integer vectors correspond through a unimodular transformation, up to a known integer offset fixed by the wrapping.*
- (b) *The covering condition (26) depends on the pairing only through the longest retained baseline, and the admissible grid grows as the cube of that length. The minimum spanning tree (MST) minimizes both over all trees.*

Proof: The proof is provided in Appendix H.

The MST is computed once from the known body frame, by Prim's or Kruskal's algorithm on the pairwise antenna distances. By (a), Theorem 2, stated for the star pairing, holds for every tree, and even the weighted estimate is the same. By (b), the pairing is a design variable of the integer layer alone. A star pairing, the tree with a single hub, is generally suboptimal. For K collinear antennas at spacing d , the chain

keeps every retained baseline at length d , while the best star retains one of length $\lceil (K-1)/2 \rceil d$ and inflates the grid by $\lceil (K-1)/2 \rceil^3$. On the regular tetrahedron all pairwise distances coincide, so every tree performs alike.

How short the retained baselines can be is set by the layout, since expanding over the centered coordinates with $\sum_k (\mathbf{b}_k - \mathbf{c}) = \mathbf{0}$ gives

$$\sum_{i < j} \|\mathbf{b}_i - \mathbf{b}_j\|^2 = K \operatorname{tr}(\mathbf{\Sigma}_{\text{eff}}). \quad (43)$$

A trace requirement therefore fixes the sum of all candidate squared lengths, and the MST is the optimal selection of $K-1$ of them from that fixed budget.

C. Aperture-Side Placement

Aperture placement reduces to one closed-form score, $\lambda_{\min}(\mathbf{\Phi})$ with $\mathbf{\Phi}$ from (38), the information along the weakest rotation axis. Under the isotropic bound (39), $\max_{\|\mathbf{a}\|=1} \mathbf{a}^T \mathbf{P}_R \mathbf{a} = 3\sigma_\varphi^2 / [\kappa_\lambda^2 \operatorname{tr}(\mathbf{\Sigma}_{\text{eff}}) \lambda_{\min}(\mathbf{\Phi})]$. Each viewing vector carries a direction and a weight. The direction $\hat{\mathbf{u}}_m \triangleq \mathbf{u}_m / \|\mathbf{u}_m\|$ bisects the illumination and reception directions, and the norm $\|\mathbf{u}_m\| = 2 \cos(\beta_m/2)$ scales the phase sensitivity, where $\beta_m \in [0, \pi)$ is the bistatic angle at the target. Hence

$$\mathbf{\Phi} = 4 \sum_{m=1}^M \cos^2(\beta_m/2) (\mathbf{I}_3 - \hat{\mathbf{u}}_m \hat{\mathbf{u}}_m^T), \quad (44)$$

which separates direction from weight. Let the M apertures share one bistatic angle β and sit symmetrically about $\hat{\mathbf{r}}_0$.

Proposition 5 (Optimal bistatic angle). *Under these conditions, $\lambda_{\min}(\mathbf{\Phi})$ is maximized at $\beta = \pi/2$, where it equals M . For $M = 2$,*

$$\lambda_{\min}(\mathbf{\Phi}) = \begin{cases} 2 \sin^2 \beta, & \beta \leq \pi/2 \\ 8 \cos^4(\beta/2), & \beta > \pi/2. \end{cases} \quad (45)$$

Minimizing $\operatorname{tr}(\mathbf{\Phi}^{-1})$ instead places the optimum at $\beta = 67.8^\circ$ for $M = 2$ and at 69.4° for every $M \geq 3$.

Proof: The proof is provided in Appendix I.

The two ends fail for opposite reasons. As $\beta \rightarrow 0$ every $\hat{\mathbf{u}}_m$ aligns with $\hat{\mathbf{r}}_0$, the $d_u = 1$ row of Corollary 1. As $\beta \rightarrow \pi$ the norm $2 \cos(\beta/2)$ vanishes, and forward scattering carries no attitude information. Between the ends the weakest axis is $\hat{\mathbf{r}}_0$, and it turns transverse only past the crossing of the two eigenvalue branches. Widening a deployment beyond $\pi/2$ therefore costs attitude accuracy, against the point-target intuition that a larger bistatic angle buys resolution. For $M \geq 3$ the score below the crossing is $M \sin^2 \beta$, so the ten-percent band $\sin^2 \beta \geq 0.9$ is the sector from 72° to 108° ; for $M = 2$ the upper end tightens to 93° . Antennas added within one site repeat the same projector in (44) and add weight without diversity, which comes from site separation, a large near-field aperture, or motion-induced viewpoints.

Table I
NUMERICAL CHECK OF THEOREM 1 AND COROLLARY 1 OVER 200 NOISE-FREE INSTANCES PER CELL.

$d_u \backslash d_b$	rank $\kappa_\lambda \mathbf{B}$			rank $\mathbf{G}_R$			solution set		
	1	2	3	1	2	3	1	2	3
1	1	2	3	1	2	2	$\mathcal{S}$	$2\mathcal{C}$	$\mathcal{C}$
2	2	4	6	2	3	3	$2\mathcal{C}^*$	2	1
3	3	6	9	2	3	3	$\mathcal{C}$	1	1

Solution sets: 1 and 2 isolated attitudes, $\mathcal{C}$ one orbit, $2\mathcal{C}$ two orbits, $\mathcal{S}$ a two-dimensional surface. Bold: $d_u + d_b \geq 5$. *199 of 200 instances, the remaining one with two orbits within the clustering radius of 0.1° .

VI. SIMULATION RESULTS

The results of Sections III-A-V are checked in two ways. Section VI-A tests the identifiability threshold, the integer layer, and the bounds against numerical geometry and the estimators. Section VI-B maps the worst-axis bound over a deployment field as stations, layout, and attitude change.

The carrier wavelength is $\lambda = 0.03$ m. The target carries $K = 8$ antennas, and every layout is centroid-referred and scaled to $\operatorname{tr}(\mathbf{\Sigma}_{\text{eff}}) = 8 \times (0.1 \text{ m})^2$, so layouts differ in shape only. The stations occupy a 60×60 m field centered at the origin, with the illuminator at one corner. The target centroid is scanned over $[-60, 60]$ m on both axes at 1 m steps. The phase-noise standard deviation of aperture m follows the two-way path loss, proportional to the product of the illuminator–target and target–aperture ranges. At the field center it averages $\sigma_\varphi = 0.03$ rad over the three apertures of Fig. 6(b). Each pixel reports the worst-axis bound $\max_{\|\mathbf{a}\|=1} (\mathbf{a}^T \mathbf{P}_R \mathbf{a})^{1/2}$ in degrees, with $\mathbf{P}_R$ the inverse of (37) at that position and attitude. Colors are logarithmic, and dark means a large bound.

A. From Bounds to Estimators

The sum condition of Theorem 1 and the catalog of Corollary 1 hold in every cell of Table I. Each cell collects 200 noise-free instances with d_u random effective viewing directions, a collinear, coplanar-ring, or cubic layout of span d_b , and a Haar-random attitude. The noise-free solution set is found by Gauss–Newton from the 2000 best nodes of the $\mathcal{SO}(3)$ grid, with the converged attitudes clustered at 0.1° . The rank of $\kappa_\lambda \mathbf{B}$ reaches 9 at (3, 3) alone, while the solution set is a single attitude in every cell with $d_u + d_b \geq 5$. At (2, 2) the rank of $\mathbf{G}_R$ is full while the solution set is a pair, so the two diagnostics of Lemma 2 part there alone. At (1, 2) and (2, 1) the solution set is two orbits rather than one, so the inclusions of the catalog are strict below the sum of four.

Fig. 3 follows that separation along one geometric parameter. Two apertures span the plane U , and a coplanar ring spans V . The attitude $\mathbf{R}_0(\vartheta) = \operatorname{Exp}(\vartheta \boldsymbol{\ell}) \mathbf{R}_{\text{ref}}$ tilts the array plane $\mathbf{R}_0 V$ out of U about their common line $\boldsymbol{\ell}$, so ϑ is the principal angle between the two planes. Two eigenvalues of $\mathbf{G}_R^T \mathbf{G}_R$ vanish as ϑ^2 , with eigenvectors within 0.04° of U , the kernel of the (2, 2)[†] row. The rows (1, 3) and (3, 1) keep exactly one zero eigenvalue along the same path. The second minimum of the noise-free cost sits at geodesic distance 2ϑ from the truth, the mirror image of the array plane in U . It

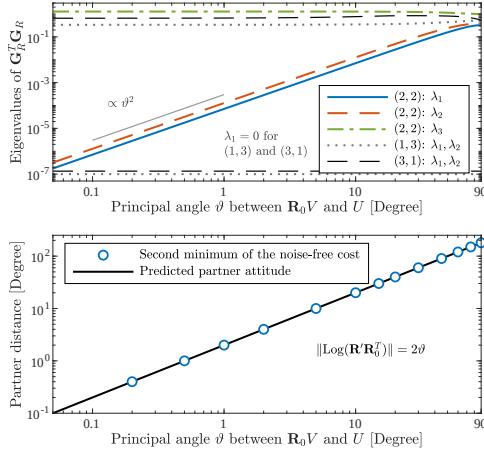


Fig. 3. The (2,2) geometry along the principal angle ϑ between the array plane $\mathbf{R}_0 V$ and the viewing plane U . Top: eigenvalues of $\mathbf{G}_R^T \mathbf{G}_R$, normalized to the largest one at 90° , with the (1,3) and (3,1) rows. Bottom: geodesic distance from the truth to the second minimum of the noise-free cost, against the partner attitude of (51).

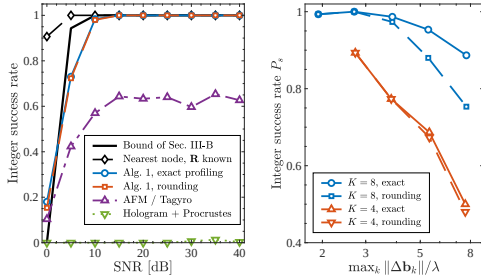


Fig. 4. Integer resolution on wrapped phases. Left: integer success rate against SNR on a grid of 5×10^5 rotations, $K = 8$, $r_T = 0.1$ m, 300 trials per point. Right: success rate of Algorithm 1 against the projected aperture at 10 dB, grid sized by (26), 150 trials per point.

coincides with (51) to 10^{-11} degrees. The partner merges with the truth exactly where the FIM turns singular.

The covering guarantee holds, and what it leaves open is the candidate selection. The remaining experiments use the stations of Fig. 6(b), the target at the field center, MST differencing, and $\sigma_\varphi^2 = 1/(2 \text{SNR})$ at the reference aperture. The left panel of Fig. 4 fixes a grid of covering radius $\delta = 2.31^\circ$, for which (26) leaves the margin $\gamma = 1.74$ rad. The union bound of Section III-B is 0.94 at 5 dB and 1 from 10 dB on. Rounding at the node nearest the true attitude succeeds in every trial from 5 dB on and in 91% at 0 dB. Algorithm 1 with $N_c = 60$ candidates reaches the same success from 15 dB on and 98% at 10 dB. With $N_c = 5$ it stalls at 95% with integer least squares and 90% with rounding at every SNR above 15 dB. The concentrated cost at the nearest node carries the grid offset, so that node does not always rank among the first five. Integer least squares stays above rounding at every point. A float-and-fix scheme does not apply, since the nine real entries of $\mathbf{R}$ and the 21 integers exceed the 21 wrapped phases. The two external methods of Fig. 4 saturate at 63% and 1.3%.

The success rate falls with the projected aperture and rises with the antenna count. The right panel fixes 10 dB and $N_c = 60$ and sizes the grid by (26) for each geometry, so

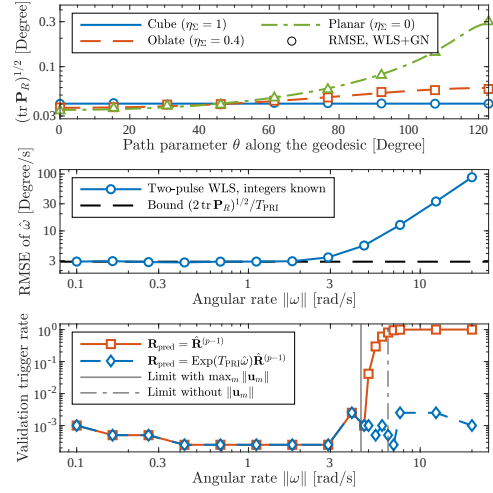


Fig. 5. Attitude dependence of the bound and the two-pulse recursion at 30 dB. Top: $(\text{tr} \mathbf{P}_R)^{1/2}$ along a geodesic for three equal-trace layouts, with WLS+GN RMSE. Middle: angular-velocity RMSE against rotation rate. Bottom: χ^2 rejection rate of the propagated integers without and with the velocity estimate.

the nearest-node margin is the same at every point. For the cube the success with integer least squares falls from 99% to 89% as $\max_k \|\Delta \mathbf{b}_k\|/\lambda$ grows from 1.9 to 7.7. Over the same range the bound $(\text{tr} \mathbf{P}_R)^{1/2}$ falls from 0.81 to 0.20. For the regular tetrahedron it falls from 89% to 50%. The gap between integer least squares and rounding widens with the aperture, to 13 points for the cube at 7.7 wavelengths. Accuracy and integer reliability therefore move oppositely with the aperture.

The bound depends on the attitude through the conjugation of Theorem 2, and isotropy removes that dependence. The top panel of Fig. 5 follows a geodesic of 123° between the attitudes at which the planar layout has its smallest and largest bound, with 500 to 2000 trials per point. The cube keeps 0.0405° along the whole path, and the closed form (39) reproduces its numerical bound to 10^{-16} . The oblate ring moves between 0.037° and 0.059° , and the planar ring between 0.035° and 0.287° . The RMSE of WLS+GN lies within 5% of the bound at all 27 attitudes.

The phase-rate estimator attains the angular-velocity bound up to the slow-rotation limit. The middle panel uses the cube, $T_{\text{PRI}} = 20$ ms, and known integers on both pulses. Up to 1.8 rad/s the RMSE stays within 3% of $(2 \text{tr} \mathbf{P}_R)^{1/2}/T_{\text{PRI}} = 2.87^\circ/\text{s}$. At 2.9 rad/s it exceeds the bound by 19%, at 4.7 rad/s by a factor of 1.9. The rotation over one interval is then 3.3° , and the second-order phase term $\kappa_\lambda \|\Delta \mathbf{b}_k\| (\|\omega\| T_{\text{PRI}})^2/2 \approx 0.08$ rad exceeds the 0.03 rad phase noise.

Integer propagation by rounding fails where the slow-rotation condition predicts, and the velocity-assisted predictor substantially extends the ambiguity-propagation limit. The bottom panel propagates the integers of a third pulse from a predicted attitude and counts the rejections of the χ_{d-3}^2 test at $\alpha = 10^{-3}$. Every rejection coincides with a wrong integer vector, and no wrong vector passes. With the previous attitude as prediction, the rejection rate stays at 0.1% up to 4 rad/s and reaches 95% at 7.0 rad/s. The onset sits at $\kappa_\lambda \max_m \|\mathbf{u}_m\| \|\omega\| T_{\text{PRI}} \max_k \|\Delta \mathbf{b}_k\| = \pi$, which gives

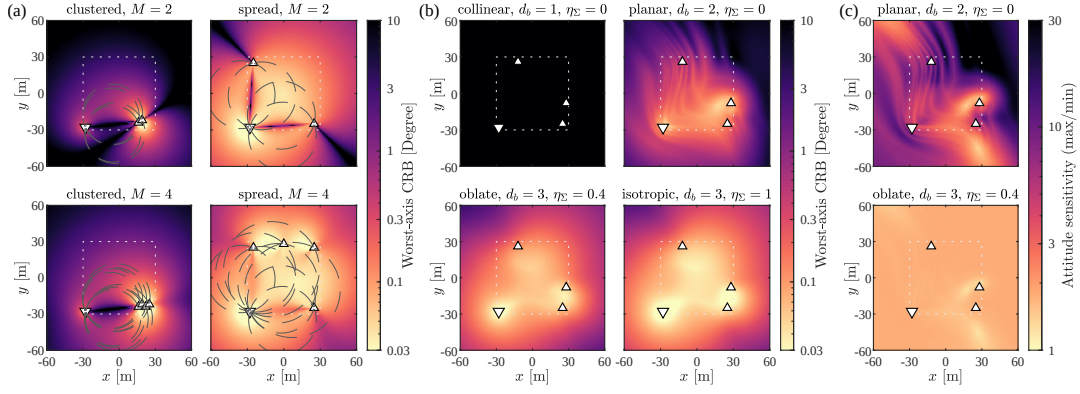


Fig. 6. Worst-axis attitude CRB over $[-60, 60]^2$ m, station field (dotted), illuminator (down-triangle), receive apertures (up-triangles). (a) Aperture-side placement: two aperture counts and two placements, coplanar stations, isotropic $K = 8$ layout, iso-bistatic-angle arcs at 72° and 108° (dashed). (b) Target-side layout: worst case over 128 attitudes for four layouts with equal $K = 8$ and $\text{tr}(\Sigma_{\text{eff}})$, three receive apertures on 6–10 m masts. (c) Ratio of the largest to the smallest worst-axis CRB over the same 128 attitudes, planar and oblate layouts of (b).

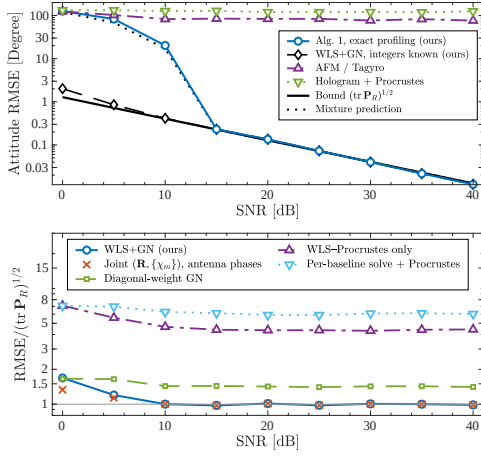


Fig. 7. End-to-end estimation from coded echoes, $K = 8$, $L = 16$, MST differencing. Top: attitude RMSE against SNR with the bound and the mixture prediction. Bottom: RMSE normalized by the bound with the integers known.

4.5 rad/s here. The same expression without $\|\mathbf{u}_m\|$ gives 6.5 rad/s, where the failure rate is already 81%. With the velocity estimate of the previous pulse pair, the prediction error replaces $\|\boldsymbol{\omega}\|T_{\text{PRI}}$ and the rejection rate stays below 0.25% up to 20 rad/s. The recursion is therefore limited by the linearization of the middle panel, not by integer propagation.

The phase-domain chain holds from the chip level up, and the integer layer sets the threshold. Fig. 7 generates every observation from (12) with the DFT code of (15), $L = 16$, and $\text{SNR} = L\rho^2/\sigma^2$ at the reference aperture, with 400 trials per point on wrapped phases and 1000 with the integers known. The empirical covariance of the differenced phases matches $\mathbf{C}_\varphi$ of Proposition 1 within 1.1% from 20 dB on. It departs by 7% at 10 dB and by 30% at 5 dB. Algorithm 1 coincides with the integers-known estimator from 15 dB on and stays within 6% of the bound. Its RMSE is 20° at 10 dB and 83° at 5 dB. Both follow the mixture prediction $\sqrt{P_s \text{tr} \mathbf{P}_R + (1 - P_s)e_w^2}$, where P_s is the success rate of Fig. 4 and $e_w = 132^\circ$ is the RMS error of the wrong-integer trials. The mixture gives 17° and 69° . The remaining gap is the lower success rate of the waveform-level phases at 5 dB, 0.64 against 0.73,

where the covariance already departs. Two external methods fit the observation model. The ambiguity-function method sums the phasors of all antenna pairs over the same grid without weighting or an inner integer minimization. It stays near 80° at every SNR. The point-target pipeline solves each baseline vector on its sphere by a hologram search and fits the attitude by Procrustes. It stays near 120° . These are the failure plateaus of Fig. 4 turned into attitude error.

Differencing loses nothing, and weighting and rigidity each carry a fixed price. The bottom panel fixes the integers and normalizes the RMSE by the bound. WLS+GN lies within 2.5% of 1 from 10 dB on. The joint estimator of $\mathbf{R}$ and $\{\chi_m\}$ on the undifferenced antenna phases returns the same attitude to 10^{-10} degrees in every trial from 10 dB on. This numerically confirms the information equivalence predicted by Proposition 2 on a trial-by-trial basis. Diagonal weights cost a factor of 1.43, or 3.1 dB, through the ignored correlation of the shared reference antenna. The WLS-Procrustes initializer alone costs 4.4, or 12.9 dB. The point-target pipeline, which uses rigidity only in its final fit, costs 6.0, or 15.6 dB. Below 10 dB all curves rise with the covariance mismatch, which marks the range of the Gaussian phase model.

B. From Bounds to Deployments

Aperture placement decides coverage, and an added aperture buys area rather than accuracy. Fig. 6(a) scans the field for two aperture counts and two placements. The isotropic layout makes each map independent of the attitude, so the aperture side is isolated. Spreading two apertures to opposite corners lowers the median bound inside the field from 1.3° to 0.08° . Two further spread apertures leave that median at 0.05° but pull its 95th percentile from 0.33° to 0.12° and halve the median outside the field. The dark bands are the two degeneracies of Theorem 1 in space. Along each illuminator-aperture segment the effective viewing direction vanishes in forward scatter. Beyond the stations the viewing directions converge to $d_u = 1$, and the path loss darkens the same region.

With the antenna count and $\text{tr}(\Sigma_{\text{eff}})$ fixed, the shape of the layout alone redraws the map. Fig. 6(b) uses three receive apertures on masts and places them unevenly, two close

together on one edge and one far away. The masts raise d_u to three, which a planar layout needs to clear the threshold. Each pixel reports the largest worst-axis bound over 128 attitudes drawn uniformly on $SO(3)$, so the maps show what a layout guarantees at every pose. The collinear layout is dark everywhere, since $d_b = 1$ leaves $d_u + d_b = 4$ below the threshold of Theorem 1. The planar layout clears the threshold but pays for the missing dimension. Its median bound is 0.9° against 0.15° for the isotropic layout, and the oblate layout with volumetricity $\eta_\Sigma = 0.4$ falls between them at 0.23° . The ordering follows Proposition 3. The trace sets the level, and the volumetricity sets how much of it survives the worst attitude.

How much the attitude matters is itself a map. Fig. 6(c) divides the largest by the smallest worst-axis bound over the same 128 attitudes for the planar and the oblate layout. For the planar layout the ratio has a median of 7 and exceeds 30 in parts of the region outside the station field. There the viewing directions converge, and a single tilt of the disc removes its baselines from all of them at once. For the oblate layout the ratio stays below 1.8 at every position, which is $[\lambda_{\max}(\Sigma_{\text{eff}})/\lambda_{\min}(\Sigma_{\text{eff}})]^{1/2}$ for this layout. A volumetric layout caps how far the pose being estimated can move the bound, and a planar layout leaves that excursion unbounded.

VII. CONCLUSION

This paper has treated a cooperative rigid body carrying a calibrated backscatter array as an encoder of its own attitude and angular velocity, and developed the identifiability theory, the intrinsic performance limits, and the design laws of that encoding for a bistatic network without carrier coherence. Three lessons carry beyond the model. Rotational sensing is governed by the viewing directions and the baselines alone, and the manifold lets one side collapse to a plane as long as the other fills space. The accuracy depends on the attitude being estimated unless the array is isotropic, so isotropy is a layout requirement. Attitude information lives in phase differences across the array, so a distributed network pays nothing for missing carrier coherence. Open questions are the validity region of the target-side far-field expansion, translation–rotation coupling within a pulse, wideband and multipath propagation, and online calibration of the array geometry.

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APPENDIX A PROOF OF LEMMA 2

Set $\mathbf{D} \triangleq \mathbf{R} - \mathbf{R}'$. By (32),

$$\begin{aligned} \mathcal{T}(\mathbf{R}) = \mathcal{T}(\mathbf{R}') &\iff \mathbf{u}_m^T \mathbf{D} \Delta \mathbf{b}_k = 0 \quad \forall (m, k) \\ &\iff \mathbf{D}(V) \subseteq U^\perp, \end{aligned} \quad (46)$$

which is statement (a). Since $(\mathbf{C}_\varphi^{(p)})^{-1} \succ \mathbf{0}$, (34) gives $\ker \mathbf{J}_R^{(p)} = \ker \mathbf{G}_R$. With $[\mathbf{G}_R \mathbf{x}]_{(m,k)} = \det[\mathbf{R}_0 \Delta \mathbf{b}_k, \mathbf{u}_m, \mathbf{x}]$, multilinearity yields $\ker \mathbf{G}_R(\mathbf{R}_0) = \mathcal{S}(\mathbf{R}_0)^\perp$ with $\mathcal{S}(\mathbf{R}_0)$ from (35), which is statement (b). ■

APPENDIX B PROOF OF THEOREM 1

Let $\mathbf{R} \neq \mathbf{R}'$ be a candidate ambiguous pair, and set $\mathbf{D} \triangleq \mathbf{R} - \mathbf{R}'$ and $\mathbf{Q} \triangleq \mathbf{R}^T \mathbf{R}' = \text{Exp}(\theta \hat{\mathbf{n}})$ with $\theta \in (0, 2\pi)$. Let $\mathbf{w} \in \text{span } U^\perp$ and $\mathbf{p} \triangleq \mathbf{R}^T \mathbf{w}$. Statements (a) and (b) refer to Lemma 2, and both are invariant under exchanging $\mathbf{R}_0 V$ with U .

1) *Spectrum of D*. The eigenvalues of $\mathbf{I}_3 - \mathbf{Q}$ are 0 and $1 - e^{\pm j\theta} \neq 0$. Hence $\text{rank } \mathbf{D} = 2$ and $\text{im } \mathbf{D} = \mathbf{R} \hat{\mathbf{n}}^\perp$, with

$$\ker \mathbf{D} = \text{span}\{\hat{\mathbf{n}}\}. \quad (47)$$

2) $d_b = 0$. $\mathcal{T}$ is constant and $\mathbf{G}_R$ is void, so $\ker \mathbf{G}_R = \mathbb{R}^3$.

3) $d_u = 1$. Let $U = \text{span}\{\hat{\mathbf{u}}\}$. Since $\text{Exp}(\theta \hat{\mathbf{u}})^T \hat{\mathbf{u}} = \hat{\mathbf{u}}$,

$$\mathbf{R}' = \text{Exp}(\theta \hat{\mathbf{u}}) \mathbf{R} \implies \mathcal{T}(\mathbf{R}') = \mathcal{T}(\mathbf{R}) \quad \forall \theta, \quad (48)$$

so (ii) fails. Differentiating (48) at $\theta = 0$ places $\hat{\mathbf{u}}$ in $\ker \mathbf{G}_R(\mathbf{R}_0)$, which also follows from $\mathcal{S}(\mathbf{R}_0) \subseteq \hat{\mathbf{u}}^\perp$ in (35). Hence (iii) fails at every $\mathbf{R}_0$. Moreover $\dim \mathcal{S} = \dim \Pi(\mathbf{R}_0 V)$ with Π the projection onto $\hat{\mathbf{u}}^\perp$, so $\ker \mathbf{G}_R = \text{span}\{\hat{\mathbf{u}}\}$ exactly when $d_b = 3$. The case $d_b = 1$ follows from the exchange invariance.

4) $d_u = d_b = 2$. Now $\dim U^\perp = 1$, so (46) forces $\dim \mathbf{D}(V) \leq 1$ and (47) gives $\hat{\mathbf{n}} \in V$. Choose a unit $\mathbf{v} \in V \cap \hat{\mathbf{n}}^\perp$, so that $V = \text{span}\{\hat{\mathbf{n}}, \mathbf{v}\}$, $\mathbf{D}\hat{\mathbf{n}} = \mathbf{0}$, and (46) reduces to $\mathbf{D}\mathbf{v} \parallel \mathbf{w}$. From $\mathbf{Q}\mathbf{v} = \cos \theta \mathbf{v} + \sin \theta (\hat{\mathbf{n}} \times \mathbf{v})$,

$$(\mathbf{I}_3 - \mathbf{Q})\mathbf{v} = 2 \sin \frac{\theta}{2} \left[\sin \frac{\theta}{2} \mathbf{v} - \cos \frac{\theta}{2} (\hat{\mathbf{n}} \times \mathbf{v}) \right]. \quad (49)$$

As θ traverses $(0, 2\pi)$ the bracket sweeps every line of $\hat{\mathbf{n}}^\perp$ except $\text{span}\{\hat{\mathbf{n}} \times \mathbf{v}\}$, hence

$$\exists \theta : \mathbf{D}\mathbf{v} \parallel \mathbf{w} \iff \mathbf{p} \perp \hat{\mathbf{n}} \text{ and } \mathbf{v}^T \mathbf{p} \neq 0. \quad (50)$$

If $\mathbf{R}V \neq U$ then $\mathbf{p} \notin V^\perp$, so $\dim(V \cap \mathbf{p}^\perp) = 1$ fixes $\hat{\mathbf{n}}$ up to sign, and $\mathbf{v}^T \mathbf{p} = 0$ would give $\mathbf{p} \perp V$. Both conditions of (50) therefore hold, with

$$\tan \frac{\theta}{2} = -\frac{\mathbf{v}^T \mathbf{p}}{(\hat{\mathbf{n}} \times \mathbf{v})^T \mathbf{p}}. \quad (51)$$

If $\mathbf{R}V = U$ then $\mathbf{p} \perp V$, so $\mathbf{D}\mathbf{v} \parallel \mathbf{w}$ forces $\mathbf{v}^T \mathbf{R}^T \mathbf{D}\mathbf{v} = \mathbf{v}^T (\mathbf{I}_3 - \mathbf{Q})\mathbf{v} = 1 - \cos \theta = 0$, impossible for $\theta \in (0, 2\pi)$. Since $\{\mathbf{R} : \mathbf{R}V = U\}$ is one-dimensional in $\mathcal{SO}(3)$, (51) holds at almost every attitude.

Turning to (35), if $\mathbf{R}_0 V = U$ then every generator of $\mathcal{S}(\mathbf{R}_0)$ lies along the normal of that common plane, so $\dim \mathcal{S} = 1$ and $\ker \mathbf{G}_R = U$. If $\mathbf{R}_0 V \neq U$, put $\text{span}\{\ell\} = \mathbf{R}_0 V \cap U$ and take $\mathbf{a} \in \mathbf{R}_0 V$, $\mathbf{u} \in U$ outside it. Then $\ell \times \mathbf{u}$ and $\mathbf{a} \times \ell$ are the two plane normals and $\text{span } \ell^\perp$, while

$$(\mathbf{a} \times \mathbf{u})^T \ell = \det[\mathbf{a}, \mathbf{u}, \ell] \neq 0, \quad (52)$$

because a vanishing determinant would force $\mathbf{u} \in \mathbf{R}_0 V \cap U = \text{span}\{\ell\}$. Hence $\mathcal{S} = \mathbb{R}^3$.

5) $d_u + d_b \geq 5$. Let $d_b = 3$. Then $\mathbf{D}(V) = \text{im } \mathbf{D}$ is two-dimensional, so (46) forces $\dim U^\perp \geq 2$, that is $d_u \leq 1$, contradicting $d_u \geq 2$. For (35), $\{\mathbf{a} \times \mathbf{u} : \mathbf{a} \in \mathbb{R}^3\} = \mathbf{u}^\perp$ holds for every $\mathbf{u} \neq \mathbf{0}$, and $\mathbf{u}_1^\perp + \mathbf{u}_2^\perp = \mathbb{R}^3$ for independent $\mathbf{u}_1, \mathbf{u}_2 \in U$. Hence $\mathcal{S}(\mathbf{R}_0) = \mathbb{R}^3$ at every attitude. The case $d_u = 3$ follows from the exchange invariance. Steps 2–5 exhaust $d_u + d_b \leq 4$ and give (i) $\Leftrightarrow$ (ii) $\Leftrightarrow$ (iii). ■

APPENDIX C PROOF OF COROLLARY 1

The entries follow from the case analysis of Appendix B. Step 2 gives the row $(\cdot, 0)$. Step 3 gives the rows $(1, \cdot)$ and $(\cdot, 1)$, together with the condition $d_u + d_b = 4$ for equality. Step 4 gives the two rows $(2, 2)$, with the partner (51) at $\mathbf{R}_0 V \neq U$ and $\ker \mathbf{G}_R(\mathbf{R}_0) = U$ at $\mathbf{R}_0 V = U$. No pair with $d_u + d_b \leq 4$ falls outside these rows, so the list is exhaustive. ■

APPENDIX D PROOF OF THEOREM 2

1) *Noise covariance*. Without interference and with $|h_{m,k}| = \rho$, (15) reduces (23) to $\mathbf{C}_\theta^{(m)} = \sigma_\varphi^2 \mathbf{I}_K$, and the star pairing $\mathbf{D} = [-\mathbf{1}_{K-1}, \mathbf{I}_{K-1}]$ gives

$$\mathbf{C}_\varphi^{(m)} = \sigma_\varphi^2 (\mathbf{I}_{K-1} + \mathbf{1}_{K-1} \mathbf{1}_{K-1}^T), \quad (53)$$

whose inverse is

$$(\mathbf{C}_\varphi^{(m)})^{-1} = \frac{1}{\sigma_\varphi^2} \left(\mathbf{I}_{K-1} - \frac{1}{K} \mathbf{1}_{K-1} \mathbf{1}_{K-1}^T \right). \quad (54)$$

2) *Per-aperture FIM*. By (33) and $\mathbf{a} \times \mathbf{u} = -[\mathbf{u}]_\times \mathbf{a}$, the rows of $\mathbf{G}_R$ that belong to aperture m stack into $-(\mathbf{u}_m]_\times \mathbf{R}_0 \mathbf{D}_\Delta)^T$, where $\mathbf{D}_\Delta \triangleq [\Delta \mathbf{b}_2, \dots, \Delta \mathbf{b}_K] \in \mathbb{R}^{3 \times (K-1)}$. The m -th block of (34) is therefore

$$\mathbf{J}_{R,m} = \kappa_\lambda^2 [\mathbf{u}_m]_\times \mathbf{R}_0 \mathbf{D}_\Delta (\mathbf{C}_\varphi^{(m)})^{-1} \mathbf{D}_\Delta^T \mathbf{R}_0^T [\mathbf{u}_m]_\times^T. \quad (55)$$

Substituting (54) and using $\Sigma = \mathbf{D}_\Delta \mathbf{D}_\Delta^T$ and $\mu = \mathbf{D}_\Delta \mathbf{1}_{K-1}$,

$$\mathbf{D}_\Delta \left(\mathbf{I}_{K-1} - \frac{1}{K} \mathbf{1}_{K-1} \mathbf{1}_{K-1}^T \right) \mathbf{D}_\Delta^T = \Sigma - \frac{\mu \mu^T}{K} = \Sigma_{\text{eff}}. \quad (56)$$

The block-diagonal $\mathbf{C}_\varphi^{(p)}$ of Proposition 1 makes (34) the sum of the M blocks, which is (37).

3) *Vectorized form*. Applying $\text{vec}(\mathbf{A} \mathbf{S} \mathbf{A}^T) = (\mathbf{A} \otimes \mathbf{A}) \text{vec}(\mathbf{S})$ to each summand with $\mathbf{A} = [\mathbf{u}_m]_\times$ and $\mathbf{S} = \mathbf{R}_0 \Sigma_{\text{eff}} \mathbf{R}_0^T$ gives the vectorized form of Theorem 2. Setting $\mathbf{S} = \mathbf{I}_3$ and using $[\mathbf{u}_m]_\times [\mathbf{u}_m]_\times^T = \|\mathbf{u}_m\|^2 \mathbf{I}_3 - \mathbf{u}_m \mathbf{u}_m^T$ gives $\Psi \text{vec}(\mathbf{I}_3) = \text{vec}(\Phi)$ with Φ from (38). ■

APPENDIX E
PROOF OF PROPOSITION 2

Let $\mathbf{G}_m \triangleq \partial \mathbf{g}_m / \partial \boldsymbol{\vartheta}^T|_{\boldsymbol{\vartheta}=\mathbf{0}}$. The first claim reads

$$\sum_{m=1}^M (\mathbf{D}\mathbf{G}_m)^T (\mathbf{D}\mathbf{C}_\theta^{(m)}\mathbf{D}^T)^{-1} (\mathbf{D}\mathbf{G}_m) = \mathbf{J}_{\vartheta\vartheta} - \mathbf{J}_{\vartheta\chi} \mathbf{J}_{\chi\chi}^{-1} \mathbf{J}_{\chi\vartheta}^T, \quad (57)$$

where $\mathbf{J}_{\vartheta\vartheta} = \sum_m \mathbf{G}_m^T (\mathbf{C}_\theta^{(m)})^{-1} \mathbf{G}_m$, $[\mathbf{J}_{\vartheta\chi}]_{:,m} = \mathbf{G}_m^T (\mathbf{C}_\theta^{(m)})^{-1} \mathbf{1}_K$, and $\mathbf{J}_{\chi\chi} = \text{diag}(\mathbf{1}_K^T (\mathbf{C}_\theta^{(m)})^{-1} \mathbf{1}_K)_{m=1}^M$ are the blocks of the joint FIM of $(\boldsymbol{\vartheta}, \chi_1, \dots, \chi_M)$. Fix one aperture and drop its index, so that $\mathbf{G} \in \mathbb{R}^{K \times 3}$, $\mathbf{C} \succ 0$, and $\mathbf{D}\mathbf{1} = \mathbf{0}$ with $\text{rank } \mathbf{D} = K - 1$.

1) *Projector identity.* For any $\mathbf{C} \succ 0$,

$$\mathbf{D}^T (\mathbf{D}\mathbf{C}\mathbf{D}^T)^{-1} \mathbf{D} = \mathbf{C}^{-1} - \frac{\mathbf{C}^{-1} \mathbf{1}\mathbf{1}^T \mathbf{C}^{-1}}{\mathbf{1}^T \mathbf{C}^{-1} \mathbf{1}} \triangleq \mathbf{M}. \quad (58)$$

Indeed, the left-hand side is invariant to $\mathbf{D} \mapsto \mathbf{S}\mathbf{D}$ for invertible $\mathbf{S}$, and any two full-rank matrices annihilating $\mathbf{1}$ differ by such an $\mathbf{S}$. Choosing in its place $\mathbf{U}^T$ with $\mathbf{U}$ an orthonormal basis of $\mathbf{1}^\perp$, both sides annihilate $\mathbf{1}$ and act as the identity on $\mathbf{C} \text{ span}(\mathbf{U})$, and $\mathbf{C} \text{ span}(\mathbf{U}) \oplus \text{span}(\mathbf{1}) = \mathbb{R}^K$ because $\mathbf{1} = \mathbf{C}\mathbf{U}\mathbf{c}$ would force $\mathbf{1}^T \mathbf{C}^{-1} \mathbf{1} = 0$.

2) *Schur complement.* The joint Fisher information of $(\boldsymbol{\vartheta}, \chi)$ has blocks $\mathbf{J}_{\vartheta\vartheta} = \mathbf{G}^T \mathbf{C}^{-1} \mathbf{G}$, $\mathbf{J}_{\vartheta\chi} = \mathbf{G}^T \mathbf{C}^{-1} \mathbf{1}$, and $\mathbf{J}_{\chi\chi} = \mathbf{1}^T \mathbf{C}^{-1} \mathbf{1}$. Its Schur complement with respect to χ is $\mathbf{G}^T \mathbf{M} \mathbf{G}$, which by (58) equals the Fisher information $(\mathbf{D}\mathbf{G})^T (\mathbf{D}\mathbf{C}\mathbf{D}^T)^{-1} (\mathbf{D}\mathbf{G})$ of the differenced data. Summing over the independent apertures gives (57), the first claim. Inverting both sides gives the second.

3) *Centroid orthogonality.* With white noise, $\mathbf{J}_{\vartheta\chi} \propto \mathbf{G}^T \mathbf{1} = -\kappa_\lambda (\mathbf{R}_0 \sum_{k=1}^K \mathbf{b}_k) \times \mathbf{u}$, which vanishes when $\sum_k \mathbf{b}_k = \mathbf{0}$. The joint FIM is then block diagonal and its Schur complement equals $\mathbf{J}_{\vartheta\vartheta}$, the FIM with known common phases, which is the third claim. ■

APPENDIX F
PROOF OF COROLLARY 2

Throughout, $\mathbf{J}_R \succ 0$ by the standing assumption of Corollary 2. Under $\|T_p \boldsymbol{\omega}^{(p)}\| \ll 1$ the nominal attitudes satisfy $\mathbf{R}^{(p)} = \text{Exp}(T_p \boldsymbol{\omega}^{(p)}) \mathbf{R}^{(p-1)}$. With $\mathbf{a} \triangleq T_p \delta \boldsymbol{\omega}^{(p)}$ and $\mathbf{b} \triangleq \boldsymbol{\vartheta}^{(p-1)}$, the Baker–Campbell–Hausdorff expansion gives $\text{Exp}(\mathbf{a}) \text{Exp}(\mathbf{b}) \approx \text{Exp}(\mathbf{a} + \mathbf{b} + \frac{1}{2} \mathbf{a} \times \mathbf{b}) \approx \text{Exp}(\mathbf{a} + \mathbf{b})$, the bracket term being second order. The kinematic constraint therefore linearizes to

$$\mathbf{R}^{(p)}(\boldsymbol{\vartheta}^{(p-1)}, \delta \boldsymbol{\omega}^{(p)}) \approx \text{Exp}(\boldsymbol{\vartheta}^{(p-1)} + T_p \delta \boldsymbol{\omega}^{(p)}) \mathbf{R}^{(p)}, \quad (59)$$

where the right-hand $\mathbf{R}^{(p)}$ is the nominal attitude. For finite nominal inter-pulse rotations the additive relation would involve the left Jacobian of $\mathcal{SO}(3)$. The first observation depends only on $\boldsymbol{\vartheta}^{(p-1)}$, whereas both perturbations enter the second through the coupled step, so with $\mathbf{G}_R \triangleq \mathbf{G}_R(\mathbf{R}^{(p-1)})$, the first Jacobian is $\mathbf{F}^{(p-1)} = [-\kappa_\lambda \mathbf{G}_R, \mathbf{0}]$ and the second is

$$\mathbf{F}^{(p)} = [-\kappa_\lambda \mathbf{G}_R, -\kappa_\lambda T_p \mathbf{G}_R]. \quad (60)$$

Since the pulse-wise phase-noise vectors are independent, the joint FIM is

$$\mathbf{J}^{(p)} = (\mathbf{F}^{(p-1)})^T \mathbf{C}_\varphi^{-1} \mathbf{F}^{(p-1)} + (\mathbf{F}^{(p)})^T \mathbf{C}_\varphi^{-1} \mathbf{F}^{(p)}. \quad (61)$$

Substituting the two Jacobians into (61), and using $\kappa_\lambda^2 \mathbf{G}_R^T \mathbf{C}_\varphi^{-1} \mathbf{G}_R = \mathbf{J}_R$, gives (40). The Schur complement of the angular-velocity block is

$$2\mathbf{J}_R - T_p \mathbf{J}_R (T_p^2 \mathbf{J}_R)^{-1} T_p \mathbf{J}_R = \mathbf{J}_R. \quad (62)$$

Thus, $\mathbf{P}_{\vartheta, \text{eff}} = \mathbf{J}_R^{-1} = \mathbf{P}_R$. The Schur complement of the attitude block is

$$T_p^2 \mathbf{J}_R - T_p \mathbf{J}_R (2\mathbf{J}_R)^{-1} T_p \mathbf{J}_R = \frac{T_p^2}{2} \mathbf{J}_R. \quad (63)$$

Therefore, $\mathbf{P}_\omega = 2\mathbf{J}_R^{-1}/T_p^2 = 2\mathbf{P}_R/T_p^2$. ■

APPENDIX G
PROOF OF PROPOSITION 3

Let $\mathbf{d} \triangleq \mathbf{b}_1 - \mathbf{c}$ and $\mathbf{T} \triangleq \sum_{k=1}^K (\mathbf{b}_k - \mathbf{c})(\mathbf{b}_k - \mathbf{c})^T$. Substitute $\Delta \mathbf{b}_k = (\mathbf{b}_k - \mathbf{c}) - \mathbf{d}$ into $\boldsymbol{\Sigma} = \sum_{k=2}^K \Delta \mathbf{b}_k \Delta \mathbf{b}_k^T$ and use $\sum_{k=2}^K (\mathbf{b}_k - \mathbf{c}) = -\mathbf{d}$. Then

$$\boldsymbol{\Sigma} = \mathbf{T} + K\mathbf{d}\mathbf{d}^T. \quad (64)$$

The same substitution gives $\boldsymbol{\mu} = -K\mathbf{d}$, so $\boldsymbol{\mu}\boldsymbol{\mu}^T/K = K\mathbf{d}\mathbf{d}^T$ and $\boldsymbol{\Sigma}_{\text{eff}} = \boldsymbol{\Sigma} - \boldsymbol{\mu}\boldsymbol{\mu}^T/K$ reduces to $\mathbf{T}$. The matrix $\mathbf{T}$ carries no reference label. Its column space is $\text{span}\{\mathbf{b}_k - \mathbf{c}\} = \text{span}\{\Delta \mathbf{b}_k\}$, since $\mathbf{b}_k - \mathbf{c} = \Delta \mathbf{b}_k - \boldsymbol{\mu}/K$, so $\text{rank}(\mathbf{T}) = d_b$. The trace bound follows from $\text{tr}(\mathbf{T}) = \sum_k \|\mathbf{b}_k - \mathbf{c}\|^2 \leq KR^2$.

For $K = 4$ collect the centered positions in the rows of $\mathbf{B} \in \mathbb{R}^{4 \times 3}$, so that $\mathbf{1}^T \mathbf{B} = \mathbf{0}$ and $\mathbf{T} = \mathbf{B}^T \mathbf{B}$. Isotropy reads $\mathbf{B}^T \mathbf{B} = \sigma^2 \mathbf{I}_3$ with $3\sigma^2 = \text{tr}(\mathbf{T})$. Then $\mathbf{B}\mathbf{B}^T$ is positive semidefinite with eigenvalues $\{\sigma^2, \sigma^2, \sigma^2, 0\}$ and null space $\text{span}\{\mathbf{1}\}$. Hence

$$\mathbf{B}\mathbf{B}^T = \sigma^2 (\mathbf{I}_4 - \frac{1}{4} \mathbf{1}\mathbf{1}^T). \quad (65)$$

Reading off $\|\mathbf{b}_i - \mathbf{b}_j\|^2 = 2\sigma^2$ for every pair shows the antennas equidistant, a regular tetrahedron. The converse follows by substitution. ■

APPENDIX H
PROOF OF PROPOSITION 4

1) *Statement (a).* Let $\mathbf{D}_T \in \{-1, 0, 1\}^{(K-1) \times K}$ stack the signed edge rows of a spanning tree T , so that $\mathbf{D}_T \mathbf{1} = \mathbf{0}$ and $\text{rank } \mathbf{D}_T = K - 1$. For trees T_1, T_2 , every edge of T_2 is the signed sum of the T_1 edges along its unique path in T_1 , so $\mathbf{D}_{T_2} = \mathbf{L}\mathbf{D}_{T_1}$ with $\mathbf{L}$ integer. The reverse expansion gives an integer $\mathbf{M}$ with $\mathbf{D}_{T_1} = \mathbf{M}\mathbf{D}_{T_2}$, and full row rank forces $\mathbf{L}\mathbf{M} = \mathbf{I}$, so $\det \mathbf{L} = \pm 1$. The unwrapped differenced phases $\tilde{\varphi} + 2\pi \mathbf{n}$ and the baselines transform by $\mathbf{L}$. Since $\mathbf{L}\tilde{\varphi}_1 = \tilde{\varphi}_2 + 2\pi \mathbf{q}$ for a known $\mathbf{q} \in \mathbb{Z}^{K-1}$, the integers transform as $\mathbf{n}_2 = \mathbf{L}\mathbf{n}_1 + \mathbf{q}$. Per aperture

$$(\mathbf{L}\mathbf{G})^T (\mathbf{L}\mathbf{C}\mathbf{L}^T)^{-1} (\mathbf{L}\mathbf{G}) = \mathbf{G}^T \mathbf{C}^{-1} \mathbf{G}, \quad (66)$$

which proves the invariance of the weighted criterion and of the FIM.

2) *Statement (b).* The bound (26) contains the pairing only through $\max_k \|\Delta \mathbf{b}_k\|$, and $|\mathcal{G}_\delta| \propto \delta^{-3}$. Let T^* be a minimum spanning tree with longest edge e , and let T be any spanning tree. Removing e splits T^* into two components, and T contains an edge f across that cut. Exchanging e for f in T^* yields a spanning tree of weight $w(T^*) - w(e) + w(f)$, so minimality gives $w(f) \geq w(e)$. The longest edge of T is therefore at least $w(e)$. ■

APPENDIX I
PROOF OF PROPOSITION 5

Write $c \triangleq \cos^2(\beta/2)$. Substituting $\|\mathbf{u}_m\| = 2\cos(\beta_m/2)$ into (38) and factoring $\|\mathbf{u}_m\|^2$ out of each summand gives (44). Take $\hat{\mathbf{r}}_0 = \mathbf{e}_z$, so that $\hat{\mathbf{u}}_m$ has polar angle $\beta/2$ and azimuth $2\pi m/M$.

1) $M \geq 3$. Equal azimuthal spacing gives

$$\sum_{m=1}^M \hat{\mathbf{u}}_m \hat{\mathbf{u}}_m^T = Mc \mathbf{e}_z \mathbf{e}_z^T + \frac{M}{2}(1-c)(\mathbf{I}_3 - \mathbf{e}_z \mathbf{e}_z^T), \quad (67)$$

so (44) has the spectrum

$$\text{eig}(\Phi) = \{M \sin^2 \beta, 2Mc(1+c), 2Mc(1+c)\}, \quad (68)$$

the first entry belonging to $\mathbf{e}_z$ and the repeated entry to the transverse plane. The first entry is the smaller one exactly when $c \geq 1/3$. It is maximized at $\beta = \pi/2$ with value M , and the repeated entry decreases in β . Hence $\lambda_{\min}(\Phi)$ is maximized at $\beta = \pi/2$, where it equals M .

2) $M = 2$. With azimuths 0 and π the cross terms cancel and (44) has the spectrum

$$\text{eig}(\Phi) = \{8c(1-c), 8c^2, 8c\} \quad (69)$$

along $\mathbf{e}_z$, $\mathbf{e}_x$ and $\mathbf{e}_y$. Since $8c^2 \leq 8c$, and $8c(1-c) \leq 8c^2 \iff \tan^2(\beta/2) \leq 1$, the minimum is as stated in (45). Both branches equal 2 at $\beta = \pi/2$.

3) *Trace criterion*. Inverting (69) and (68),

$$\text{tr}(\Phi^{-1}) = \begin{cases} \frac{1+c-c^2}{8c^2(1-c)}, & M=2 \\ \frac{5-3c}{12c(1-c^2)}, & M=3. \end{cases} \quad (70)$$

The numerators of their derivatives in c are $-(c^3-2c^2-2c+2)$ and $-(6c^3-15c^2+5)$. Their roots in $(0,1)$ are $c = 0.6889$ and $c = 0.6759$, that is $\beta = 67.8^\circ$ and $\beta = 69.4^\circ$.

The second angle serves every $M \geq 3$. Inverting (68) gives

$$\text{tr}(\Phi^{-1}) = \frac{1}{M} \left[\frac{1}{4c(1-c)} + \frac{1}{c(1+c)} \right],$$

in which M appears only as a common factor, so the stationary angle does not depend on it. ■

APPENDIX J
PROOF OF PROPOSITION 1

Fix aperture m and pulse p , and drop the pulse index. Let $\mathbf{v}_m \triangleq \boldsymbol{\xi}_m + \mathbf{n}_m \in \mathbb{C}^L$ be the chip-domain perturbation, with covariance $\mathbf{R}_v^{(m)} = \mathbf{R}_\xi^{(m)} + \sigma^2 \mathbf{I}_L$.

1) *Matched filtering and gain calibration*. Operations 1) and 2) of Section III-A map $\mathbf{v}_m$ to

$$\bar{\mathbf{v}}_m = \frac{1}{L} \mathbf{H}_m^{-1} \mathbf{C}_{\text{cf}}^H \mathbf{v}_m, \quad (71)$$

where $\bar{\mathbf{v}}_m = [\bar{v}_{m,1}, \dots, \bar{v}_{m,K}]^T$, and $\mathbf{C}_{\text{cf}}$ and $\mathbf{H}_m$ are as in (23). Hence

$$\mathbf{Q}_m \triangleq \text{Cov}(\bar{\mathbf{v}}_m) = \frac{1}{L^2} \mathbf{H}_m^{-1} \mathbf{C}_{\text{cf}}^H \mathbf{R}_v^{(m)} \mathbf{C}_{\text{cf}} \mathbf{H}_m^{-H}. \quad (72)$$

By (15) the thermal part of $\mathbf{Q}_m$ is the diagonal $(\sigma^2/L) \mathbf{H}_m^{-1} \mathbf{H}_m^{-H}$, and only the interference part couples the columns.

2) *Phase linearization*. The calibrated noiseless entry of column k is $e^{-j(\phi_{m,k}^{\text{tot}} - \psi_m)}$. For small $\bar{v}_{m,k}$,

$$\theta_{m,k} \approx \text{Im}\{\bar{v}_{m,k} e^{j(\phi_{m,k}^{\text{tot}} - \psi_m)}\}, \quad (73)$$

where $\theta_{m,k}$ is the phase noise of column k before differencing. Stacking over k gives $\boldsymbol{\theta}_m \approx \text{Im}\{e^{-j\psi_m} \mathbf{S}_m \bar{\mathbf{v}}_m\}$. For proper-complex $\mathbf{z}$, $\text{Cov}(\text{Im}\{\mathbf{z}\}) = \frac{1}{2} \text{Re}\{\text{Cov}(\mathbf{z})\}$, and the unit scalar $e^{-j\psi_m}$ leaves the covariance unchanged, so

$$\mathbf{C}_\theta^{(m)} = \frac{1}{2} \text{Re}\{\mathbf{S}_m \mathbf{Q}_m \mathbf{S}_m^H\}, \quad (74)$$

which is (23).

3) *Differencing*. The pairing gives $\boldsymbol{\epsilon}_m = \mathbf{D} \boldsymbol{\theta}_m$, hence $\mathbf{C}_\varphi^{(m)} = \mathbf{D} \mathbf{C}_\theta^{(m)} \mathbf{D}^T$. Perturbations uncorrelated across apertures make the stacked covariance block diagonal, which is (22). ■